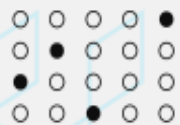




CLINOMA PLATFORM

ophthalmology boards



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Tip: Use the interactive landscape view on your tablet or laptop for optimal clinical visualization.

CHAPTER 01

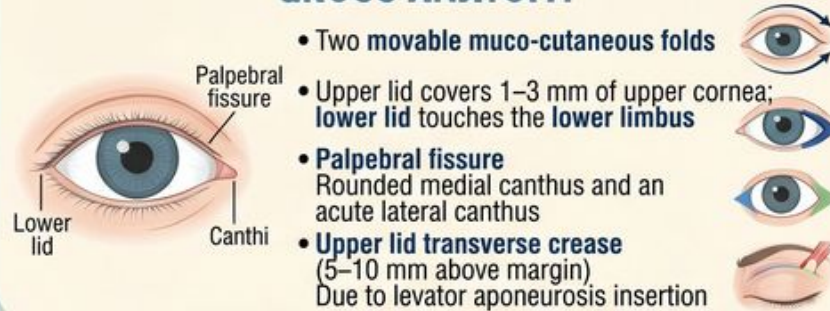
Chapter 1: Eyelid



COMPREHENSIVE ANATOMY & PHYSIOLOGY OF THE EYELIDS • BLEPHARITIS (CHRONIC INFLAMMATION OF EYELID MARGIN) • ACUTE SUPPURATIVE INFLAMMATIONS OF EYELID GLANDS (STYE & HORDEOLUM INTERNUM) • CHALAZION (CHRONIC LIPO-GRANULOMA) & DISTICHIASIS • ACQUIRED MISDIRECTED LASHES (TRICHIASIS & RUBBING LASHES) • ENTROPION & ECTROPION (LID MARGIN MALPOSITIONS) • LAGOPHTHALMOS (INCOMPLETE LID CLOSURE) • PTOSIS (DROOPING OF UPPER EYELID)

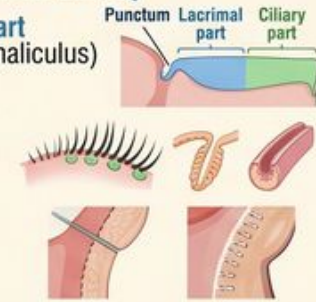
COMPREHENSIVE ANATOMY & PHYSIOLOGY OF THE EYELIDS

GROSS ANATOMY



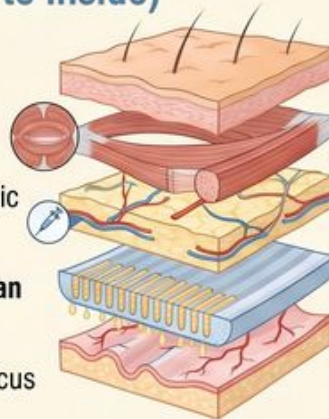
LID MARGIN (2–3 mm wide)

- Divided by the **punctum** into a **Lacrimal part** (medial 1/6th, no cilia/glands, contains canaliculus) and a **Ciliary part** (lateral 5/6th)
- **Ciliary part** has:
 - **Anterior border** (cilia, Zeiss' sebaceous glands, Moll's sweat glands)
 - **Grey line** (for surgical splitting)
 - **White line** (Meibomian gland orifices)
 - **Sharp posterior border**



LAYERS (Outside to Inside)

- 1) **Skin & Subcutaneous tissue**
Thin, elastic, no fat
- 2) **Muscle layer**
Striated and non-striated muscles
- 3) **Submuscular areolar tissue**
Contains nerves and arteries (anesthetic injection site deep to orbicularis)
- 4) **Fibrous layer (Tarsus)**
Forms the skeleton; embeds **Meibomian glands** which secrete an oily film
- 5) **Palpebral conjunctiva**
Vascular, firmly adherent, contains sulcus sub tarsalis (2 mm above margin)



EYELID MUSCLES & FUNCTIONS

• Eyelid Muscles

- 1) **Orbicularis Oculi (7th Nerve)**
 - **Orbital part** (tight closure)
 - **Palpebral part** (gentle closure, contains Riolan's muscle)
 - **Lacrimal part** (Horner's muscle) for lacrimal pump
 - **Paralysis** causes **lagophthalmos** and **ectropion**
- 2) **Levator Palpebrae Superioris (3rd Nerve)**
Arises at orbit apex, inserts into superior tarsus, skin crease, and upper fornix
 - **Paralysis** causes **ptosis**
- 3) **Involuntary Muscles**
 - **Muller's muscle** (sympathetic supply; paralysis causes **Horner's syndrome**)
 - **Lower lid retractors** (attached to inferior tarsus)

• Functions

- Globe protection
- Tear distribution (blinking)
- Tear drainage (lacrimal pump)
- Aqueous drainage (ocular massage)

BLEPHARITIS (Chronic Inflammation of Eyelid Margin)



DEFINITION Chronic inflammation of the eyelid margin.

MAIN TYPES



1) Anterior Blepharitis

Affects skin around eyelashes.



Squamous

(seborrheic/metabolic, associated with scalp dandruff, shows **greasy grayish-white scales**)



Ulcerative

(caused by *Staphylococcus aureus*, shows lashes glued by **yellow crusts**; removal leaves **bleeding ulcers**)



2) Posterior Blepharitis

Affects Meibomian glands, causing **Meibomian gland dysfunction (MGD)**.



3) Mixed Blepharitis

Combination of both anterior and posterior blepharitis.

CLINICAL PICTURE



Burning sensation



Itching



Lacrimation



Photophobia



Frequent falling of hair

COMPLICATIONS



- **Lacrimal**
Punctal occlusion.



- **Conjunctiva**
Chronic conjunctivitis.



Marginal corneal ulcers and superficial punctate keratitis.



- **Eyelid**
Trichiasis (scarring at roots), madarosis (destruction of follicles), multiple styes, and destruction of the sharp posterior border leading to **epiphora**, initiating a vicious circle of **epiphora-eczema-ectropion**.

TREATMENT



Warm compresses



Removal of scales using diluted baby shampoo

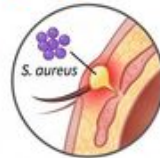


Local antibiotic ointment

Acute Suppurative Inflammations of Eyelid Glands (Stye & Hordeolum Internum)

Hordeolum Externum (Stye)

Definition: Acute suppurative inflammation of the **Zeiss gland** caused by ***Staphylococcus aureus***.



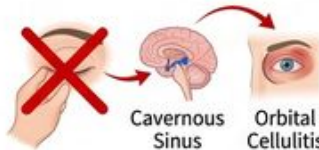
Clinical Picture: Painful red swelling at the **eyelid margin**, forming a **yellow point or abscess** at the root of a lash.



Treatment: Warm compresses, topical antibiotics (drops and ointment), and systemic antibiotics for severe cases with cellulitis. If pointing, **drain pus by epilation** of the related lash with a horizontal incision.



Contraindication: Do not press to evacuate a stye; it may **spread infection** causing **orbital cellulitis** or **cavernous sinus thrombosis**.



Hordeolum Internum

Definition: Acute suppurative inflammation of the **Meibomian gland**. Can be **primary**, or **secondary** due to **infection** of a **chalazion** (acute chalazion).



Etiology: Causative organisms are the same as a stye.

 = Stye
S. aureus

Clinical Picture: Same as a stye, but **pain is more severe** because the gland is larger and **embedded in the tarsus under tension**. The **swelling is away from the lid margin** and not related to eyelashes. A **yellow spot appears on the conjunctival side**.



Treatment: Similar to stye, but **evacuation** may be needed through a **vertical or cruciate incision** on the conjunctival side.



EYE CONDITIONS: CHALAZION & DISTICHIASIS

CHALAZION (Chronic Lipo-granuloma)

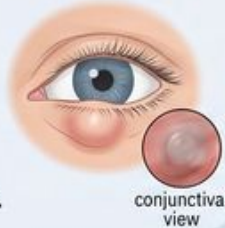
Definition

Localized **chronic inflammatory lipo-granuloma** of a Meibomian gland.



Clinical Picture

Painless swelling under the skin of the eyelid. Pain occurs if it becomes **infected (acute chalazion)**. Presents as a localized, firm, painless, non-tender swelling with **freely mobile skin** over it. The palpebral conjunctiva looks red or **bluish-grey** over it.



Treatment

Large chalazion requires **incision (cruciate or vertical)** and **curettage/scraping** from the conjunctival side. Infected chalazion is treated as hordeolum internum.

Recurrent swelling in the same gland requires **excision biopsy** and **histopathological examination** to exclude malignancy.



DISTICHIASIS

Definition

Congenital abnormality where an **extra row of lashes** arises from the lid margin behind the grey line at the **Meibomian gland orifices**.



Treatment

Destruction of ectopic lashes by selective destruction or **cryotherapy** on the anterior surface of the tarsus after **splitting the lid** at the grey line.



ACQUIRED MISDIRECTED LASHES (TRICHIASIS & RUBBING LASHES)

RUBBING LASHES vs. TRICHIASIS



Rubbing Lashes refers to **4** or less inward misdirected lashes rubbing against the cornea or bulbar conjunctiva.



Trichiasis is an acquired condition in which **MORE THAN 4** lashes are directed posteriorly and rub against the globe.

CAUSES of ACQUIRED MAL-DIRECTED LASHES

Cicatrizing diseases affecting the lid margin:



Trachoma (commonest in Egypt)



ulcerative blepharitis



trauma



and chemical burns

CLINICAL PICTURE

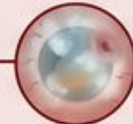


Symptoms:
Foreign body sensation, pain, lacrimation, photophobia, and conjunctival hyperemia.



Signs:
Misdirected lashes with/without entropion, conjunctival hyperemia, and **corneal vascularization** in long-standing cases.

COMPLICATIONS



Conjunctival chronic conjunctivitis/ulceration, **corneal recurrent ulceration**, **opacification**, and epithelial plaques.

TREATMENT LINES



Temporary: Epilation (lashes regrow within 2–3 weeks).



Permanent: Destruction of involved lashes by electrolysis, argon laser ablation, or diathermy.

ENTROPION & ECTROPION: LID MARGIN MALPOSITIONS

Common Eyelid Misalignments

ENTROPION

Entropion: Rolling in of the lid margin



• Etiological Types:

-  **Congenital** (common in lower lid)
-  **Cicatricial** (caused by trachoma, burns, shortens posterior lamella)
-  **Spastic** (spasm of Riolan's muscle from acute inflammation)
-  **Senile/Involutional** (lower lid of elderly due to elastic tissue degeneration)

• Clinical Picture:

-  Same as acquired mal-directed lashes

ECTROPION

Ectropion: Rolling out (outward rotation) of the lid margin

• Degrees:

-  **Mild** (punctum visible)
-  **Moderate** (tarsal conjunctiva exposed)
-  **Severe** (completely everted lid exposing fornix)

• Etiological Types:

-  **Congenital** (lid laxity)
-  **Cicatricial** (skin scars from burns/trauma)
-  **Senile/Involutional** (most common, lower lid due to)
-  **Mechanical** (tumor/everted lid exposing fornix)
-  **Lagophthalmos** with exposure keratopathy in severe paralytic cases

• Clinical Picture:

-  **Epiphora** (causing eczema and skin scarring, leading to a vicious circle)
-  **Chronic conjunctival hyperemia**, dryness, hypertrophy
-  **Lagophthalmos** with exposure keratopathy in severe paralytic cases
-  **Mechanical** (tumor/chalazia weight)
-  **Paralytic** (7th nerve Bell's palsy)
- Treatment of Paralytic Ectropion:**
 -  **Help nerve regeneration** (Vitamin B, vasodilators, cortisone)
 -  **Help muscle** (massage, electrical stimulation)
 -  **Protect cornea** (artificial tears by day, ointment at night, lid strapping)

LAGOPHTHALMOS (INCOMPLETE LID CLOSURE)

Explanatory Guide & Key Points

• DEFINITION:



Incomplete closure of the **palpebral fissure** on voluntary or involuntary closure.

• BELL'S PHENOMENON:

A **protective mechanism** where **rolling up** of the eye during **sleep** **diminishes** the exposure effect.



• DEGREES:



Mild (can close lids voluntarily by forcible squeezing, but remain open during sleep).



Severe (palpebral fissure does not close even with forcible squeezing).

• ETIOLOGY:



- **Local Physiological:** Occurs during sleep (especially in children) due to lack of orbicularis tone.



- **Local Pathological (Lid lesions):** Severe ectropion, structural loss (coloboma), overcorrection of ptosis/entropion, spastic lid retraction (thyroid ophthalmopathy), and LMN facial palsy.



- **Local Pathological (Globe lesions):** Proptosis, anterior staphyloma, buphthalmos.



- **General Causes:** Extreme debility, unconsciousness, and general anesthesia.

• COMPLICATIONS:



Exposure keratitis (may end in **corneal ulceration, perforation**, and **eye loss**) and **dry, xerotic conjunctiva**.

• TREATMENT:



Aimed at **protecting the cornea**.

Temporary lines include **ointment** at night, narrow **plaster strips** for closure, or a **contact lens**.

Surgical lines include **lateral tarsorrhaphy** (reduces fissure width) or **median tarsorrhaphy** (union of whole lid margin in severe cases).

Always **remove the primary cause**.

PTOSIS: Drooping of Upper Eyelid

(Normally covers the upper 1/6th of cornea)

• DEFINITION & ETIOLOGY

• A. Congenital:



Faulty development of levator muscle or its nerve, dated since birth. Can be associated with weak superior rectus or **Marcus Gunn phenomenon** (Jaw Winking, where the lid elevates on jaw movement).



• B. Acquired:



• **Neurogenic** (3rd nerve palsy, Horner syndrome)



• **Myogenic** (trauma, tumor, dystrophy, Myasthenia Gravis [periodic, worst in evening, relieved by anticholinesterase])



• **Aponeurotic** (Senile degeneration)



• **Traumatic**



• **Mechanical** (increased lid weight from tumor/chalazia) or **Hysterical**.

• CLINICAL PICTURE & COMPLICATIONS



Drooping lid, absent lid crease. Compensatory frontalis contraction (forehead wrinkling) and chin elevation.

Complications:

- Ocular torticollis
- Amblyopia
- Squint.



• CLINICAL GRADING

- Amount of Ptosis (MRD):

Mild (1–2 mm)



Moderate (3 mm)



Severe (> 4 mm)



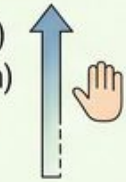
- Levator Function Power:

Normal (≥ 15 mm)

Good (11–14 mm)

Fair (5–11 mm)

Poor (≤ 4 mm)



• TREATMENT



Acquired requires treating the **underlying cause** (wait 6 months for paralytic/traumatic).



Congenital mild/moderate can wait until **school age** (5–6 years), but **severe** must be operated early to avoid amblyopia.



• CHOICE OF OPERATION

If levator action is fair or good, do **Levator resection and advancement**.



If levator function is poor & severe ptosis, do **Frontalis suspension (sling) / Hiss operation**.



CHAPTER 02

Chapter 2: Lacrimal System



ANATOMY OF THE LACRIMAL SECRETORY SYSTEM & TEAR FILM • ANATOMY & PHYSIOLOGY OF THE LACRIMAL DRAINAGE SYSTEM • WATERING OF THE EYE (LACRIMATION VS. EPIPHORA) • CLINICAL EVALUATION AND INVESTIGATIONS OF EPIPHORA • SURGICAL MANAGEMENT OF EPIPHORA • ACUTE AND CHRONIC DACRYOCYSTITIS • DRY EYE SYNDROME & LACRIMAL GLAND DISEASES

ANATOMY OF THE LACRIMAL SECRETORY SYSTEM & TEAR FILM

The Lacrimal Secretory System consists of:

1) The **main lacrimal gland**.



2) The **accessory lacrimal glands**: Microscopic in size, opening into the conjunctiva by fine ductules.

Includes **Krause's glands** (located in upper and lower fornices)



Wolfring glands (located in the palpebral conjunctiva opposite mid-tarsus).

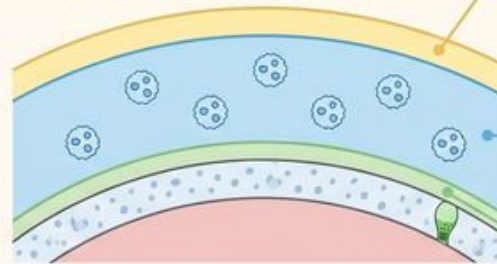


The Tears:

A **watery isotonic** fluid with a slightly **alkaline pH**.



PRECORNEAL TEAR FILM (Three Layers):



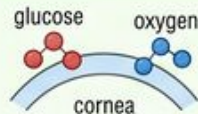
1) **Outer lipid layer**: Secreted by Meibomian and accessory Zeiss glands. It prevents rapid tear **evaporation**, prevents tear **overflow**, and **lubricates** eyelid movement.



2) **Middle aqueous layer**: Secreted by main and accessory lacrimal glands. It forms the **main bulk** of tears, cleans and **flushes** debris, and supplies **oxygen** and **antibacterials** like **lysozymes** to the corneal epithelium.

3) **Inner mucin layer**: Secreted by conjunctival goblet cells. It **covers the hydrophobic corneal** epithelium with a **hydrophilic surface** for the aqueous layer to spread on.

Functions of Tears:



Nutrition (supplies **glucose** and **oxygen** to the cornea)



Optical (maintains **uniformity** by filling epithelial irregularities)



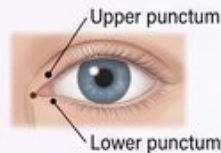
Protection (**wetting** against dryness, mechanical **washout**, and **antimicrobial** action)

Anatomy & Physiology of the Lacrimal Drainage System

Components of the Drainage System

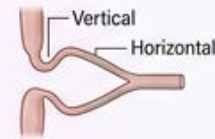
> 1) Two lacrimal puncta

Upper and lower openings located at the **posterior edge** of the lid margin at the **junction** of the lacrimal and ciliary parts.



> 2) Two canaliculi

Fine tubes carrying tears from puncta to the sac. Each has a short **vertical part** and a **longer horizontal part**, opening separately or uniting into a **common canaliculus**.



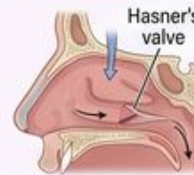
> 3) Lacrimal sac

Lies in the **lacrimal fossa** in the medial orbital wall. It is surrounded by **Horner's muscle** which is the **pars lacrimalis** of orbicularis oculi.



> 4) Nasolacrimal duct

Ends in the **inferior meatus** of the nose, guarded by **Hasner's valve** which acts as a **one-way valve** allowing tears down but preventing air from going up.



Tear Drainage Mechanism



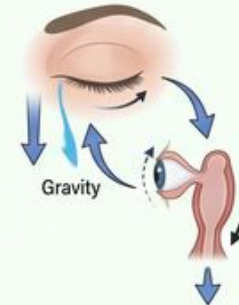
- Evaporation

Accounts for a **small percentage** of tear loss.



- Excretion

Occurs via **Passive forces** like **gravity** and **capillarity**, and **Active forces** which include the **pump mechanism** of the lacrimal sac driven by **Horner's muscle** during blinking.



Watering of the Eye (Lacrimation vs. Epiphora)

Definition:  Overflow of tears on the cheek. It is divided into:



Lacrimation:

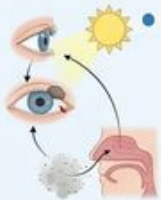
Over-secretion of tears with normal, patent lacrimal passages.



- **Causes:** Psychic and emotional conditions



- **Synkinesis** with cranial nerves during yawning or vomiting



- **Reflexes** such as bright light via the optic nerve, trigeminal irritation like a corneal foreign body, or nasal mucosa irritation by dust and smoke



Epiphora:

Obstruction of the lacrimal passages with normal, baseline tear secretion.



- Causes due to **failure of pump mechanism:** Irregular sharp lid margin, ectropion, or orbicularis muscle paralysis in facial nerve palsy



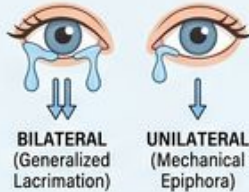
- Causes due to **structural obstruction:** Congenital absence, stenosis, or scarring of puncta and canaliculi from post-trachomatous changes, fungal infections, herpes, or trauma; dacryocystitis, fractures, or tumors of the lacrimal sac; imperforated Hasner's valve, local vein congestion, or nasal pathology such as polypi, deviated septum, and turbinate hypertrophy

Clinical Evaluation and Investigations of Epiphora

A comprehensive guide to diagnosing watering eyes.

• HISTORY:

- **Bilateral watering** usually indicates generalized lacrimation;
- **unilateral watering** strongly points to mechanical epiphora.



• CLINICAL EXAMINATION:

- Check eyelids to exclude **ectropion or trichiasis**,
- look for visible **lacrimal sac swelling** or inflammation,
- and inspect the nose for **polypi or a deviated septum**.



• REGURGITATION TEST:

- **Technique:** Apply firm pressure over the lacrimal sac against the lacrimal bone.
- **Positive Regurgitation:** Reflux of **pus, mucoid fluid, or tears** from the puncta, indicating **nasolacrimal duct obstruction**. This serves as a **definite proof of anatomical blockage**.
- **Negative Regurgitation:** **No fluid reflux**, signifying **patent** passages or a **blockage higher up** in the canaliculi.



• IMAGING AND ENT EVALUATION:

- **Dacryocystography:** Visualizes **anatomical details** and **exact site of strictures** in the drainage system using radiopaque contrast material.
- **Computed Tomography scan.**
- **ENT Examination:** Vital to discover **external outlet obstructions** and evaluate the **status of the nasal mucosa**.



SURGICAL MANAGEMENT OF EPIPHORA

Explanatory Guide to Procedures

GENERAL APPROACH



Treatment is primarily aimed at the **underlying cause**, such as the **surgical correction of ectropion**.

FOR PUNCTAL & CANALICULAR STENOSIS OR OBSTRUCTION



- **Dilatation and probing.**



- **One-snip ampulectomy:** Making a **vertical snip** in the **posterior wall** of the canaliculus.



- **Laser punctoplasty.**



- **Two-snip operation:** A **small triangle of tissue** is removed from the **posterior wall** of the canaliculus to keep it open.



- **Conjunctivo-dacryocystorhinostomy (CDCR):** Connecting the conjunctiva directly to the nose using a **specialized glass tube** to **bypass severe canalicular strictures**.

FOR NASOLACRIMAL DUCT OBSTRUCTION (NLDO)

CONGENITAL OBSTRUCTION



- **Stepwise approach** starting with **medical hydrostatic massage** over the sac and **topical antibiotics** during the **first year** of life.



- Followed by **mechanical dilatation and probing** if it fails.



- If probing fails repeatedly, **silastic intubation** or a **surgical bypass** is performed.

ACQUIRED OBSTRUCTION



- Probing usually **fails in adults**.



- Requires a **Dacryocystorhinostomy (DCR)** to create a **permanent fistula** between the sac and the **nasal mucosa**.



- OR a **Dacryocystectomy** if the **bypass is contraindicated**.



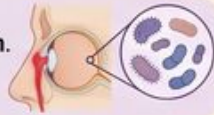
ACUTE & CHRONIC DACRYOCYSTITIS: UNDERSTANDING LACRIMAL SAC INFLAMMATION

ACUTE DACRYOCYSTITIS

Acute suppurative inflammation of the lacrimal sac.

Etiology & Predisposition

- Predisposed by underlying **nasolacrimal duct obstruction**.
- Common causative organisms include **Pneumococci**, **Staphylococci**, and **Streptococci**.

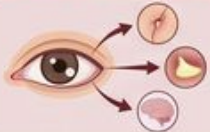


Clinical Picture



- **Severe throbbing pain** over the sac area.
- **Persistent epiphora** (watering).
- Fever, headache, and a **tender red swelling** that may show **abscess formation or fluctuation**.

Complications



- **Lacrimal fistula opening** through the skin.
- **Pyocele** (pus-filled cavity).
- Progression to **chronic dacryocystitis**.
- **Orbital cellulitis** and **cavernous sinus thrombosis** (serious).

Treatment



- **Systemic and local antibiotics**, hot fomentation, and **analgesics**.
- If a fluctuant abscess forms, **incision and drainage** are required.
- For a permanent fistula, a **fistulectomy** with a **dacryocystorhinostomy** is performed.

CHRONIC DACRYOCYSTITIS

Chronic lacrimal sac inflammation secondary to persistent nasolacrimal duct obstruction.

Etiology & Predisposition

- **Pneumococcus** is the isolated agent in the vast majority of cases.

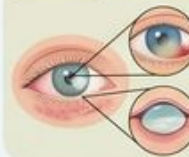


Clinical Picture



- **Epiphora** accompanied by **mucoid or purulent discharge**.
- A **hyperemic inner canthus**.
- A painless sac swelling below the medial palpebral ligament.
- A positive regurgitation test (pressure yields discharge).

Complications



- Dangerous source of infection leading to **hypopyon** and **corneal ulcers** from minor abrasions.
- **Eczema** or **ectropion** from chronic watering.
- **Recurrent conjunctivitis**.
- **Mucocele**.
- Major **septic contraindication** for any **intraocular surgery**.

Treatment



- **Dacryocystorhinostomy** to connect the sac to the middle meatus nasal mucosa.
- **Dacryocystectomy** is strictly reserved for **tuberculosis**, tumors, extreme sac fibrosis, or cases with an **active hypopyon ulcer** where **urgent eradication of infection** is needed.



Dacryocystectomy

- Removal of the sac
- tuberculosis, or specific conditions

DRY EYE SYNDROME & LACRIMAL GLAND DISEASES



DRY EYE SYNDROME

Partial or absolute deficiency in aqueous tear production or stability.

CAUSES:



Congenital alacrima;
surgical removal or **trauma**
to the gland;



chronic sarcoidosis
dacryoadenitis;



Keratoconjunctivitis sicca;
mucin deficiency from
severe conjunctival scarring
in trachoma or chemical burns;



vitamin A deficiency;



lagophthalmos or **facial**
nerve palsy;



senile glandular atrophy;



and **iatrogenic drugs** like
anti-anxiety medications.

CLINICAL PICTURE:



Ocular irritation, foreign body
and burning sensation;



a **deficient or absent tear**
film with a scanty lower
lid tear strip;



punctate epithelial erosions,
filamentary keratitis;
wrinkling, and **Bitot's spots** in
nutritional deficiencies.

TREATMENT:



Usage of **protective glasses,**
frequent **artificial tear drops**
and **lubricating gels,**



punctal occlusion to retain tears,



therapeutic **soft bandage contact lenses,**



mucolytics like **acetylcysteine,** and short courses
of **local steroids.**



Measured quantitatively
via **Schirmer's test,**



qualitatively via
Tear break-up time,



and **surface staining**
using **Rose Bengal dye.**



DISEASES OF THE LACRIMAL GLAND

TYPES:



Dacryoadenitis which
is an acute or chronic
inflammation,



Atrophy of the gland
leading to dry eye
status,



and **Lacrimal gland**
tumors which range from
simple mixed adenoma
to highly malignant
tumors.

CHAPTER 03

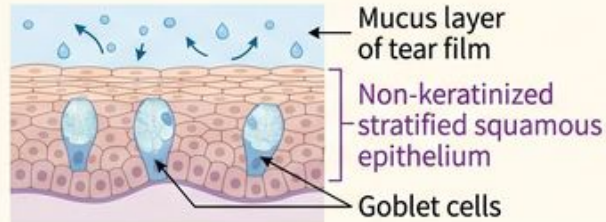
Chapter 3: Conjunctiva



NORMAL ANATOMY & HISTOLOGY OF THE CONJUNCTIVA • ACUTE MUCOPURULENT CONJUNCTIVITIS (MPC) • ACUTE PURULENT CONJUNCTIVITIS (PC / ACUTE BLENORRHOEA) • OPTHALMIA NEONATORUM • INCLUSION & ADENOVIRAL CONJUNCTIVITIS • TRACHOMA (EGYPTIAN OPTHALMIA) - PATHOGENESIS & MCCALLAN'S CLASSIFICATION • WHO CLASSIFICATION & CORNEAL MANIFESTATIONS / COMPLICATIONS OF TRACHOMA • VERNAL KERATO-CONJUNCTIVITIS (VKC / SPRING CATARRH) • PHLYCTENULAR KERATO-CONJUNCTIVITIS & CONJUNCTIVAL DEGENERATIONS • SYMBLEPHARON

Normal Anatomy & Histology of the Conjunctiva

- Histologically, the conjunctiva consists of two primary layers:



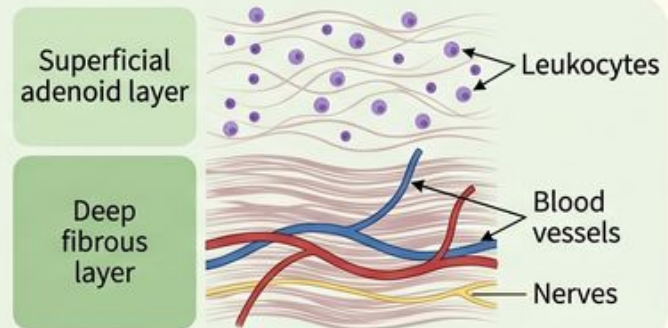
1) The Epithelium

Composed of **non-keratinized stratified squamous epithelium**. It contains specialized **goblet cells** that secrete mucus to form the inner layer of the tear film.

2) The Substantia Propria

Composed of two distinct layers:

- **Superficial adenoid layer**: Formed of loose connective tissue containing numerous leukocytes.
- **Deep fibrous layer**: Much denser in structure, containing the blood vessels and nerves of the conjunctiva.



- Clinical Classification of Conjunctival Diseases:**



Inflammations
(Conjunctivitis)



Allergic
reactions



Degenerations

ACUTE MUCOPURULENT CONJUNCTIVITIS (MPC)

DEFINITION

An acute infective conjunctivitis characterized by **MARKED HYPEREMIA** and **MUCOPURULENT DISCHARGE**.



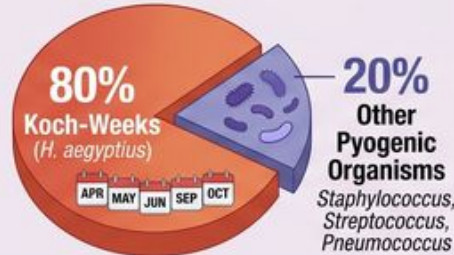
PREDISPOSING FACTORS

- **BAD HYGIENE,**
- **POOR SANITATION,** and
- **A CONTAMINATED ENVIRONMENT.**



CAUSATIVE ORGANISMS

- **KOCH-WEEKS BACILLUS** (80% of cases, also known as *Haemophilus aegyptius*). Highly common in Egypt during warm seasonal months (April, May, June, September, and October).



CLINICAL PICTURE

SYMPTOMS:
DISCOMFORT,
RED EYE, and
ACTIVE DISCHARGE.



SIGNS:
EDEMA of the **EYELID,**
MARKED CONJUNCTIVAL
HYPEREMIA (most prominent
at the fornices),
CONJUNCTIVAL
EDEMA (CHEMOSIS), and
MUCOPURULENT DISCHARGE
(pus and mucus) that
characteristically glues the
eyelashes together in the
morning due to overnight
accumulation.



COMPLICATIONS

RARE, but may include
SUPERFICIAL MARGINAL
CRESCENTIC CORNEAL
ULCERS and **TOXIC**
IRITIS (especially with
pneumococci).



TREATMENT

- **LOCAL ANTIBIOTIC EYE DROPS AND OINTMENT,**
- **HOT FOMENTATION** to induce vasodilation (brings antibodies and washes toxins), and
- **DARK GLASSES** to avoid photophobia.



ABSOLUTE CONTRAINDICATION



EYELID BANDAGE is
COMPLETELY CONTRAINDICATED
as it prevents the free exit of
discharge, increases the
temperature of the
conjunctival sac, and promotes
BACTERIAL MULTIPLICATION
and **TOXIN RETENTION.**

ACUTE PURULENT CONJUNCTIVITIS (PC / ACUTE BLENORRHOEA)

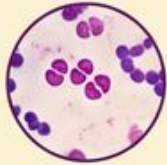
DEFINITION

An acute, **HIGHLY CONTAGIOUS** suppurative inflammation of the conjunctiva characterized by **SEVERE HYPEREMIA, COPIOUS PURULENT DISCHARGE, PREAURICULAR LYMPHADENOPATHY**, and a **STRONG TENDENCY FOR CORNEAL INVOLVEMENT**.



CAUSATIVE ORGANISMS

GONOCOCCUS (80%, Gram-negative diplococci) and **OTHER BACTERIA** (20%, e.g., *Streptococci*, *Staphylococci*).



COMPLICATIONS

SEVERE MARGINAL CORNEAL ULCERS (can coalesce to form a **RING ULCER**) or **CENTRAL/PARACENTRAL ULCERS** that **COMMONLY PERFORATE**. This occurs because Gonococci can invade an **INTACT CORNEAL EPITHELIUM**, and **SEVERE CHEMOSIS** compresses limbal capillaries (impairing nutrition) while forming a physical gutter that traps proteolytic enzymes and exotoxins. Can also cause **SPASTIC ECTROPION** and **TOXIC IRIDOCYCLITIS**.



TRANSMISSION & TYPES

Epidemic type: Occurs primarily in **SUMMER** (July–August).

Genital type: Caused by **AUTOINFECTION** via hands from genital gonorrhea (more common in males) or from maternal passages during **LABOR** (*Ophthalmia neonatorum*).



CLINICAL PROGRESSION (STAGES)

1) STAGE OF INFILTRATION (2–3 days): MARKED **EYELID EDEMA** closing the palpebral fissure, severe conjunctival **CHEMOSIS**, SUBCONJUNCTIVAL HEMORRHAGE, watery/mucoid discharge, and mild fever with **ENLARGED, TENDER PREAURICULAR LYMPH NODES**.



2) STAGE OF DISCHARGE (2–3 weeks): DIMINISHING EDEMA but development of **PROFUSE, THICK PURULENT DISCHARGE**.



3) CHRONIC STAGE (2–3 months): SUBSIDING EDEMA, redness localized to fornices, **MINIMAL DISCHARGE** but full of **GONOCOCCI** (carrier state).



TREATMENT

SYSTEMIC and **ENERGETIC LOCAL BROAD-SPECTRUM ANTIBIOTICS** (DROPS AND OINTMENT), combined with **MYDRIATIC-CYCLOPLEGIC DROPS** if the **CORNEA IS INVOLVED**.



OPHTHALMIA NEONATORUM

Explanatory Guide: Definition, Causes, Diagnosis, Features, & Treatment



• DEFINITION:

Any form of **ACUTE CONJUNCTIVITIS** occurring within the **FIRST MONTH** after delivery. Transmitted from maternal passages during labor or via contaminated towels.



• ETIOLOGY (Causes):



Infectious:
GONOCOCCI (50%), **CHLAMYDIA TRACHOMATIS** (Inclusion blenorrhoea), **OTHER BACTERIA** (Staph, Strept, B. coli;ococcus), and **HERPES SIMPLEX VIRUS**.



Non-infectious:
CHEMICAL CONJUNCTIVITIS induced by irritation from **PROPHYLACTIC DROPS** (silver nitrate, savlon).



• DIAGNOSTIC NOTE:



The **LACRIMAL GLAND** does not secrete tears until at least **TWO WEEKS** after birth; hence, **ANY DISCHARGE** during the **FIRST WEEK** in a neonate must be considered highly suspicious.



• CLINICAL FEATURES BY AGENT:



CHEMICAL: ONSET WITHIN HOURS. Self-limited, **MILD SEROUS DISCHARGE**, lasts 24–36 hours.



N. GONORRHOEAE:

ONSET at 24–48 **HOURS**. Hyperacute, **COPIOUS PURULENT DISCHARGE**, severe lid swelling/chemosis. **REQUIRES** systemic Benzyl penicillin (50,000 u/kg for 7 days) **OR** single IM Cefotaxime (100 mg/kg) **PLUS** topical tobramycin/fluoroquinolone drops.



CHLAMYDIA TRACHOMATIS:

ONSET at 5–12 **DAYS**. Mild to moderate **PURULENT DISCHARGE**, marked redness of palpebral conjunctiva. **TREATED** with local terramycin ointment **AND** systemic erythromycin (50 mg/kg/day for 3 weeks).



NON-GONOCOCCAL BACTERIA: ONSET at 5–30 **DAYS**. **VARIABLE SEVERITY**.



HERPES SIMPLEX VIRUS:

ONSET WITHIN 2 WEEKS. Mild injection, **SEROSANGUINOUS DISCHARGE**, occasional vesicular skin rash. **TREATED** with anti-herpetic drops/ointments.



• PROPHYLAXIS & TREATMENT:



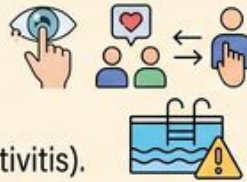
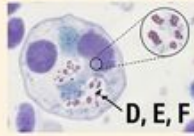
PREVENTION includes **TREATMENT** of the **MOTHER** before labor, **WASHING THE NEWBORN** from above downwards, and utilizing **CREDE'S METHOD** (prophylactic antibiotic eye drops). **CURATIVE TREATMENT** mimics purulent conjunctivitis but requires **URGENT CARE** due to the **THIN NEONATAL CORNEA** and lack of protective lysozymES.



INCLUSION & ADENOVIRAL CONJUNCTIVITIS

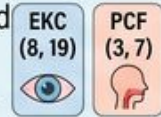
• INCLUSION CONJUNCTIVITIS (Adult/Neonatal)

- **Etiology:** Caused by *Chlamydia oculogenitalis* (Serotypes D, E, F).
- **Transmission:** Auto-infection from the genitalia via **fingers** in young adults with **chlamydial urethritis/cervicitis**, **sexual transmission**, or through **contaminated swimming pools** (swimming pool conjunctivitis).
- **Signs:** **Hyperemia**, **mucopurulent discharge**, **large follicles** located mainly in the **lower palpebral conjunctiva**, **superficial keratitis** (without pannus), and **preauricular lymphadenopathy**.
- **Treatment** mirrors the **medical therapy for trachoma**.



• ADENOVIRAL CONJUNCTIVITIS

- **Etiology & Transmission:** **Highly contagious** via **finger-to-eye contact**; **self-limited** (5–14 days).
- **Types:** **Epidemic Keratoconjunctivitis (EKC)**, caused by adenovirus **types 8 and 19** and **Pharyngoconjunctival fever (PCF)**, caused by **types 3 and 7**.
- **Clinical Picture:** **Acute red eye**, **photophobia**, **lacrimation**, **conjunctival follicles** (predominantly in the **lower conjunctiva and fornix**), **subconjunctival hemorrhages**, and **preauricular lymphadenopathy**.
- **Key Differences:** **EKC** is characterized by **progressive keratitis** (punctate epithelial keratitis -> subepithelial lesions -> stromal infiltrates). **PCF** is characterized by **systemic symptoms like fever and pharyngitis**.
- **Treatment:** Supportive (cold compresses, lubricants, dark glasses). **Antivirals are ineffective**. **Topical steroids** do not shorten the course but **suppress severe corneal inflammation** if vision drops.



TRACHOMA (EGYPTIAN OPHTHALMIA) – Pathogenesis & McCallan's Classification

Understanding the causes, progression, and clinical stages

PATHOGENESIS & TRANSMISSION

• DEFINITION

A chronic **infective keratoconjunctivitis** caused by *Chlamydia trachomatis*, characterized by the formation of **follicles, papillae, and pannus**, healing exclusively by **cicatrization (scarring)** leading to potential **lid deformities and blindness**.



• CAUSATIVE AGENT & TRANSMISSION

Chlamydia trachomatis serotypes **A, B, Ba, and C** are associated with endemic trachoma. It is transmitted via **direct/indirect contact** with infected discharge (**hands, clothes, towels**), with **flies** playing a primary role.



McCALLAN'S CLINICAL STAGES

• 1) T1 (Trachoma at onset)

Incubation of 6–12 days. **Gritty sensation, lid heaviness, mucoid discharge, and photophobia**. Shows small, immature, non-raised, grayish, non-expressible **follicles** on the **upper tarsal conjunctiva**, separated by areas of vasodilation. Early upper **corneal active infiltration** (Trachomatous pannus) begins.



• 2) T2 (Established or florid trachoma)

Presence of well-developed **mature soft follicles** and intense **papillary hypertrophy**.

• **T2a:** Dominated by **large, mature soft follicles**.

• **T2b:** Dominated by **trachomatous papillae** (T2b' shows fine, non-uniform papillary hypertrophy; T2b'' shows flat-topped papillae mixed with spring catarrh elements).

• **T2c:** Trachoma complicated by secondary **acute gonococcal infection**.

• **T2v:** Characterized by **plasma cell infiltration** and hyaline degeneration of the conjunctiva (**Stellwag's brawny edema**).



• 3) T3 (Starting cicatrization)

Active inflammation coexists with early scarring. Shows **star-shaped or radiating white scars** on the upper tarsal conjunctiva, including **Arlt's line** (a white line of scar tissue involving the sulcus subtarsalis). May show **Post-trachomatous degenerations (PTDs)**: yellowish-white trapped epithelial/mucoid spots) and **Post-trachomatous concretions (PTCs)**: chalky white, raised, hard calcified debris resulting from calcification of PTDs).



• 4) T4 (Cicatrized or healed trachoma)

Active inflammation (follicles/papillae) is **completely absent**; only **quiet scar tissue** remains. The disease is **no longer infectious**.



WHO Classification & Corneal Manifestations / Complications of Trachoma

WHO Simplified Grading System

- TF (Trachomatous Follicles)**

Graded from F1 (trivial/absent) to F2 (mild), F3 (moderate), and F4 (severe).



- TI (Trachomatous Intense)**

Diffuse inflammatory thickening of the upper palpebral conjunctiva obscuring more than 50% of the normal deep tarsal vessels.



- TS (Trachomatous Scarring)**

Scarring in the conjunctiva (Grades 1, 2, 3).



- TT (Trachomatous Trichiasis)**

At least one in-turning eyelash rubbing the globe, or a history of epilation.



- CO (Corneal Opacity)**

Visible corneal opacity obscuring part of the pupillary margin, reducing vision to 6/18 or worse.



Corneal Manifestations

- Concomitant with conjunctival infection, selectively affecting the **upper cornea** (warmer environment under the lid).



- Trachomatous Pannus**

Cellular infiltration and superficial vascularization of the upper cornea. In early stages, infiltration lies between the epithelium and Bowman's membrane; later, Bowman's membrane is destroyed, leaving a permanent opacity (**nebula ex-pannus**) with a serrated edge.



- Herbert's Rosettes**

Grayish small elevated lymphoid follicles at the edge of the active pannus. They heal to leave permanent, depressed clear pits at the limbus called **Herbert's pits**.



- Clinical Types of Pannus**

Thin (tenuis), Vascular (vasculosus), Fleishy (carnosus), Annular (annulosus, all around cornea), and Healed/Cicatrizated (siccus).



- Course** Progressive (infiltration extends beyond vessels) vs. **Regressive** (vessels extend beyond infiltration).



- Trachomatous Corneal Ulcer**

Linear, superficial ulcer with a horizontal long axis situated at the lower border of the pannus; can also occur mechanically from from trichiasis or hard PTCs.



Complications of Trachoma

- Eyelid**

Trichiasis, cicatricial entropion (more common in upper lid), mechanical/infiltrative ptosis (infiltration of Muller's muscle).



- Conjunctiva**

Xerosis (severe dryness due to destruction of goblet cells and accessory lacrimal glands), posterior symblepharon (shallow fornix), and hyaline degeneration of tarsus/conjunctiva.



- Cornea**

Diffuse corneal nebulae leading to irregular astigmatism, dense leucoma, and corneal xerosis.



- Lacrimal**

Chronic dacryocystitis and **chronic dacryoadenitis**.



Surgical Treatment Guidelines



Expressing large follicles in **T2a**, scraping severe papillae with a sharp spoon in **T2b**, and picking/evacuating hard PTDs/PTCs in **T3** and **T4**.

VERNAL KERATO-CONJUNCTIVITIS (VKC) / SPRING CATARRH

1. DEFINITION & ETIOLOGY (Seasonal Allergic Conjunctivitis)

A **chronic, bilateral, recurrent** seasonal allergic conjunctivitis.

Driven by **IgE-mediated hypersensitivity** to *unknown exogenous allergens* (ultraviolet rays, heat, dust, pollen grains).

Primarily affects **young boys** and worsens during the **warm summer season** (spring is a misnomer).



2. CLINICAL PICTURE: Symptoms & Manifestations

SYMPTOMS

- Intense **itching**, lacrimation, photophobia, **red eye**.
- Characteristic **ropy, thick, thready mucoid discharge** rich in **eosinophils**.



CONJUNCTIVAL MANIFESTATIONS

- **Palpebral type (70%)**: Upper palpebral conjunctiva shows **large, flat-topped, dark red or bluish-white papillae** arranged in a characteristic 'cobble-stone' pattern. Upper fornix is **strictly free**.
- **Bulbar / Limbal type (10%)**: Shows **gelatinous masses** at the **limbus** (medially, laterally, or circumferentially). Contains **small, white elevated spots** called '**Tranta's spots**', formed of concretions of calcium, eosinophils, and necrotic epithelium.
- **Mixed type (20%)**: Combines **both palpebral and bulbar signs**.



CORNEAL MANIFESTATIONS

- **Punctate epithelial keratitis**: Scattered **white epithelial spots** known as *keratitis superficialis vernalis* of Tobgy.
- Epithelial **macro-erosions** forming a **sterile 'Shield ulcer'**.
- **Peripheral corneal opacities** resembling *arcus senilis*.



3. MANAGEMENT & TREATMENT

LOCAL TREATMENT

- **Washing with cold lotions**: **Cold saline** or **3% sodium bicarbonate** to dissolve discharge.
- **Topical anti-histamines**, mast cell stabilizers, **vasoconstrictors/decongestants**, and topical **NSAIDs**.



STEROIDS

- **Topical steroid drops/ointments** are highly effective for **acute flares**.
- **Prolonged use MUST be avoided** to prevent **steroid-induced glaucoma** and **complicated cataract**.



RESISTANT CASES

- **Topical immunosuppressants** (**Cyclosporine 2%**).
- **Intralesional triamcinolone steroid injections** into the **upper tarsus**.



PHLYCTENULAR KERATO-CONJUNCTIVITIS

• Definition

An acute allergic nodular condition of the conjunctiva and cornea resulting from hypersensitivity to an **endogenous allergen/toxin** (Tuberculo-protein, intestinal parasites, or chronic septic foci like streptococcal tonsillitis or staphylococcal blepharitis). It mostly affects children aged 5–12 years.

• Signs

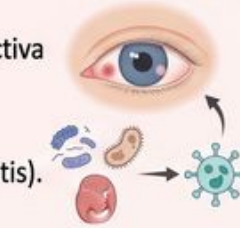
The characteristic lesion is the **Phlycten**—a round, small (1–3 mm), grayish elevated nodule surrounded by a localized zone of conjunctival injection, commonly located at the limbus or bulbar conjunctiva. It can ulcerate and suppurate under secondary infection.

• Corneal signs

Single marginal phlyctenular ulcers, a circumferential ring ulcer (fusion of multiple limbal ulcers), and **phlyctenular pannus** (vascularization straight and thin, extending deep to Bowman's membrane with a rounded cellular border).

• Treatment

General eradication of the endogenous cause, combined with topical steroids which provide a dramatic therapeutic effect, and local antibiotics.



CONJUNCTIVAL DEGENERATIONS

1) Pterygium

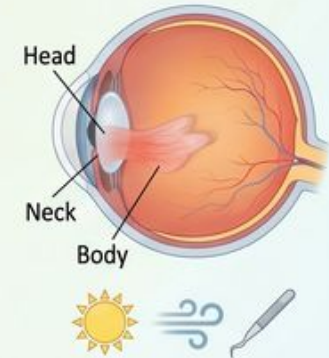
A triangular vascular encroachment of the conjunctiva onto the cornea, highly common in Egypt due to chronic exposure to dust, wind, fumes, and UV rays.

Consists of a blunt head (apex on cornea), neck (at limbus), and body (on sclera).

Located nasally and usually bilateral.

Surgery is indicated if it is progressive, encroaches on the pupil, causes disfigurement, or induces diplopia.

Operations include **simple excision** (high recurrence) or excision with bare sclera combined with a conjunctival/amniotic graft, postoperative Beta irradiation, or intraoperative Mitomycin C to prevent recurrence.



2) Pinguecula

A small, yellow, stable degenerative nodule on the nasal side of the limbus in elderly individuals, induced by UV rays.

It never suppurates, never ulcerates, and requires no treatment unless cosmetically problematic.



SYMBLEPHARON: An Explanatory Guide

Pathological Ocular Adhesions

DEFINITION & CAUSES

DEFINITION

Pathological adhesion of the **palpebral conjunctiva** to the **bulbar conjunctiva** or the **corneal surface**.



CAUSES

Any **severe ocular insult** resulting in two opposing raw, **denuded epithelial surfaces**.

- 1) Severe **cicatricial trachoma**. 
- 2) Chemical/thermal **trauma** (acid or alkali burns). 
- 3) Postoperative **surgical complications**. 
- 4) **Membranous conjunctivitis**. 
- 5) **Autoimmune blistering diseases** (Stevens-Johnson syndrome, ocular cicatricial pemphigoid). 

COMPLICATIONS & MANAGEMENT

COMPLICATIONS

-  **Diplopia** (severe limitation of ocular motility).
-  **Direct visual impairment** (scarring encroaches on cornea).
-  **Facial disfigurement**.
-  Positional lid margin anomalies (**entropion/ectropion**), and **lagophthalmos** leading to **exposure keratopathy**.

MANAGEMENT

PROPHYLACTIC (Acute phase)

-  Frequent **sweeping of the fornices** with a **glass rod** coated with **antibiotic ointment**.
-  **Topical cortisone** to check active **fibroblastic proliferation**.
-  Insertion of a **symblepharon contact shell** until complete **epithelialization**.

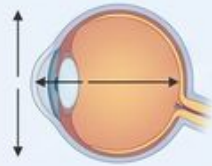
CURATIVE (Chronic phase)

-  **Surgical excision** of the symblepharon band with or without a **mucous membrane** or **amniotic membrane graft**.
-  **Keratoplasty** (corneal grafting) is required if a **dense corneal opacity** coexists.

Chapter 4: Cornea & Sclera

KERATOREFRACTIVE SURGERY & DISEASES OF THE SCLERA

Comprehensive Anatomy, Physiology, & Transparency of the Cornea



Macro-anatomy & Dimensions

The cornea forms the **anterior 1/6th** of the outer coat of the globe. Its **vertical diameter is 11 mm**, while its **horizontal diameter is 12 mm**. It has a **powerful refractive index of 1.37** and a **refractive power of 42 diopters**. It is **completely avascular** except for the peripheral 1 mm supplied by the circumcorneal ciliary plexus.

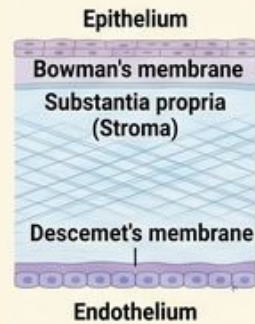


The Limbus (Corneoscleral Junction) Histological Changes



The corneal epithelium becomes continuous with the conjunctival epithelium; **Bowman's membrane ends**; the **stroma becomes continuous with the sclera**; **Descemet's membrane becomes continuous with the trabecular meshwork**; and the **endothelium continues over the angle of the anterior chamber and iris**.

Micro-anatomy (Five Layers from Outside Inwards)



- 1) Stratified squamous non-keratinized epithelium.
- 2) An elastic sheet with low resistance; when injured, it **does not regenerate**.
- 3) Represents **90% of corneal thickness**. It is composed of a uniform collagenous ground substance rich in mucopolysaccharides. Its main features are that it is **transparent and avascular**.
- 4) The **highly elastic basement membrane** of endothelial cells, showing **high resistance**; when injured, it **can regenerate**.
- 5) A single layer of flat hexagonal cells that perform an **active barrier/pump function**.

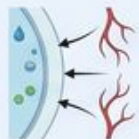


Nerve Supply



Richly supplied by the **sensory circumcorneal ciliary plexus** formed by the **two long ciliary nerves** (branches of the nasociliary nerve from the ophthalmic division of the **Trigeminal nerve**).

Physiology & Nutrition



The cornea is **hydrated**. It derives **glucose** by **diffusion** from the **aqueous humor**, **oxygen and NaCl** from the **tear film**, and **peripheral factors** from **limbal capillaries**. Its **optical function** provides the main **refractive medium** of the eye, while its **sensory nerves** prompt **quick protective lid closure**.

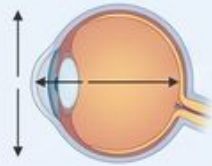


Causes of Transparency



Anatomical factors (non-keratinized regular epithelium, parallel surfaces, unmyelinated nerve fibers, and a uniform refractive index) combined with **physiological factors** (active endothelial dehydration pump).

Comprehensive Anatomy, Physiology, & Transparency of the Cornea



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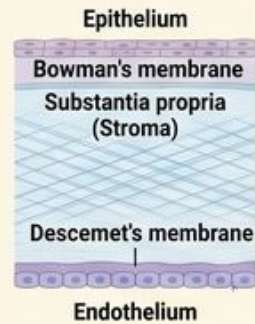


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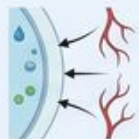


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Causes of Transparency



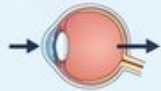
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CLASSIFICATION OF KERATITIS & HYPOPYON CORNEAL ULCER (Acute Serpiginous Ulcer)

• Classification of Keratitis

(Inflammation of the Cornea)

• By Cause:



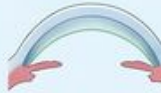
Infective (Exogenous from conjunctival sac; Endogenous which is rare; or Spread from nearby tissues like conjunctivitis, uveitis, scleritis) & **Non-Infective** (Traumatic, Allergic, or Degenerative like neurotrophic).

• By Nature:



Suppurative & Non-Suppurative.

• By Site:



Superficial & Deep.

• Hypopyon Corneal Ulcer (Acute Serpiginous Ulcer)

• Definition:



A **primary infective, deeply progressive suppurative corneal ulcer** characterized by **severe toxic iridocyclitis** and early formation of **hypopyon**.

- **Causative Organism:** **Pneumococcus** is the commonest agent, responsible for 80% of typical cases.



- **Predisposing Factors:** General **decreased resistance** (old age, diabetes mellitus, malnutrition) combined with **local epithelial damage** (trauma by fingernail, foreign body, trichiasis, PTDs), as *Pneumococcus* cannot invade an intact corneal epithelium.



- **Precipitating Factors:** Chronic dacryocystitis, conjunctivitis, blepharitis, or scratching with dirty fingers.



• Clinical Picture (Symptoms):



Stitching **pain** from exposed nerve endings, **defective vision**, **photophobia**, reflex lacrimation, blepharospasm, and **halos** around lights caused by corneal edema.

- **Clinical Signs:** Eyelid **edema**, **ciliary injection**, loss of corneal luster, and **grayish infiltration** with epithelial sloughing (confirmed by a positive green fluorescein stain test). It characteristically appears as a **paracentral disc-shaped ulcer** with an **advancing crescentic undermined edge** that creeps towards the center, preceded by cellular infiltration, while its peripheral healing edge is flat and vascularized.



- **Hypopyon Characteristics:** **Sterile pus** (composed of living and dead polymorphonuclear leukocytes, fibrin, and iris pigment) settling at the bottom of the anterior chamber with a **flat upper level**, formed by exudation from **severe toxic iridocyclitis**.



- **Complications of Suppurative Corneal Ulcers:** Secondary **toxic iridocyclitis**, secondary **glaucoma** (due to hypopyon or peripheral anterior synechiae), **descemetocoele** (Descemet's membrane bulging as a translucent bleb under normal IOP), **permanent corneal opacities**, and **sudden perforation**.



- **Perforation of Corneal Ulcer:** Occurs when deep progression is met with a sudden rise in intraocular pressure due to straining, coughing, crying, sneezing, or uncooperative eye examinations.

- **Sequelae of Corneal Perforation:** Iris prolapse, anterior synechia, leucoma adherent, lost anterior chamber with complicated cataract, corneal fistula, and endophthalmitis.



- **Corneal Fistula:** Formed when the epithelium grows over the edges of a perforated ulcer, preventing spontaneous healing. It presents with a **positive river green sign** (fluorescein leak), soft ocular tension (hypotony), a **lost anterior chamber**, **papilledema**, **anterior polar cataract**, and **peripheral anterior synechiae**.



Types of Corneal Scars (Grading after Healing):

1) Nebula (Cloud): A faint corneal scar following the healing of superficial ulcers; opacity occupies less than half of corneal thickness, and iris details are visible through it.



2) Macula (Spot): A localized scar intermediate in density between a nebula and a leucoma; opacity occupies half to two-thirds of corneal thickness, **blurring iris details**.



3) Leucoma (White): A dense, **white corneal scar** following the healing of deep ulcers; opacity is greater than half of corneal thickness, and **iris details cannot be made out**.



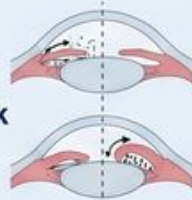
4) Adherent Leucoma: A dense **leucomatous opacity** with **incarceration and trapping of the iris tissue** within the scar.



Complications and Sequelae of Irdocyclitis

SECONDARY GLAUCOMA

Can occur via **dual mechanisms**. In the **early acute phase** (Open-angle), it is due to **mechanical blockage** of the **trabecular meshwork** by inflammatory cells, thick fibrin, or high-protein plasmoid aqueous.



In the **late chronic phase** (Closed-angle), it is due to organized **Peripheral Anterior Synechiae (PAS)**, or a complete 360-degree **Ring Posterior Synechia (Seclusio Pupillae)**.

Seclusio pupillae physically blocks aqueous flow, causing the peripheral iris to bow forward (**Iris Bombe**), leading to bow forward (**Iris Bombe**), leading to severe **secondary angle-closure glaucoma**.

CYCLITIC MEMBRANE

The **organization of rich inflammatory exudates** behind the lens into a **dense contractile fibrous membrane**, which can pull on the ciliary body causing **ciliary detachment** and **profound hypotony**.



Ciliary Detachment & Hypotony

COMPLICATED CATARACT

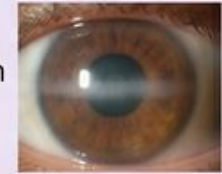
A classic **posterior subcapsular cataract** developing due to **altered aqueous metabolic nutrition**, **diffusing inflammatory toxins**, and the **prolonged therapeutic use of topical corticosteroids**.



Posterior Subcapsular Cataract

BAND KERATOPATHY

A **progressive horizontal band of grayish-white calcium salt deposition** in **Bowman's membrane** of the cornea, typically occurring in **chronic smoldering uveitis**, especially in children with **Juvenile Idiopathic Arthritis (JIA)**.



Calcium Salt Deposition in Bowman's Membrane

MACULAR COMPLICATIONS

Cystoid Macular Edema (CME) and **macular ischemia**, which stand as the **major leading causes of permanent visual loss** in chronic uveitis patients.



Cystoid Macular Edema & Ischemia

PHTHISIS BULBI

The **devastating end-stage of severe chronic destructive uveitis**; it presents as a **shrunken, totally disorganized, soft, and blind eye** due to **permanent failure of aqueous secretion** by the destroyed, fibrosed ciliary body.



Shrunken, Disorganized, Blind Eye

COMPREHENSIVE TREATMENT PROTOCOLS FOR CORNEAL ULCERS

MANAGEMENT OF UNCOMPLICATED ULCERS



Topical Antibiotics: Dual therapy (combining two antibiotics to cover both Gram-positive and Gram-negative pyogenic organisms) or Monotherapy using potent fluoroquinolones.



Cycloplegics: Atropine sulfate 1% drops or ointment to paralyze the sphincter pupillae and ciliary muscles, relieving painful spasm and dilating the pupil to keep the iris away from the lens to avoid posterior synechiae.



Bilateral Bandage: Stops eyelid movement, cutting down mechanical irritation of exposed nerve endings to relieve pain and preventing the peeling of new epithelium to promote rapid healing.

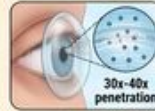


Hot Fomentation: Wet or dry compresses act as a counter-irritant to relieve pain, improve local circulation, wash away toxins, and bring immunoglobulins and leukocytes.

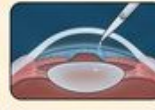
MANAGEMENT OF RESISTANT ULCERS



Subconjunctival antibiotic injections to bypass the epithelial barrier and achieve high local concentrations.



Collagen shields (dissolvable high-dose drug delivery contact lenses) to increase antibiotic penetration 30-to-40-fold.



Paracentesis (surgical puncture and evacuation of the anterior chamber using a special knife).



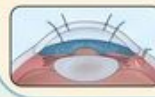
Amniotic membrane graft to provide substrate material covering the bed.



Conjunctival flap (covering the ulcer with vascular bulbar conjunctiva to provide direct defense and nutrition).



Tarsorrhaphy (lateral or median to protect the cornea in exposure states).

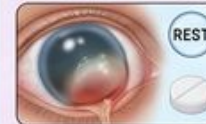


Therapeutic lamellar or penetrating keratoplasty.

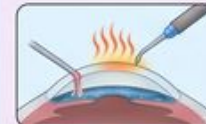
MANAGEMENT OF COMPLICATIONS



Secondary Glaucoma: Antiglaucoma drops and repeated paracentesis.



Descemetocoele & Perforation: Absolute bed rest, avoiding straining, oral Diamox to lower IOP, or urgent therapeutic keratoplasty.



Corneal Fistula: Chauffage (indirect thermal cautery close to the fistula to destroy its epithelial lining); if it fails, therapeutic keratoplasty is performed.



Corneal Opacities: Corrected using contact lenses or optical keratoplasty.

VIRAL KERATITIS: Dendritic Ulcer (Herpes Simplex Virus)

DEFINITION

Primary infective corneal ulcer, **linear, superficial, crenated or dendritic** in shape.



ETIOLOGY

Caused by **Herpes Simplex Virus (HSV)**, epitheliotropic. **No solid immunity.**



INFECTION STAGES

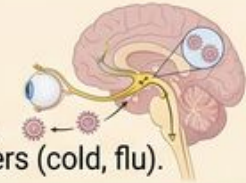
Primary Infection

Occurs in children; severe **follicular conjunctivitis** with fever, lymph node enlargement, corneal involvement.



Latency & Recurrence

Virus dormant in Gasserian ganglion (Trigeminal nerve). Reactivated by local heat, immunosuppression (steroids), systemic fevers (cold, flu).



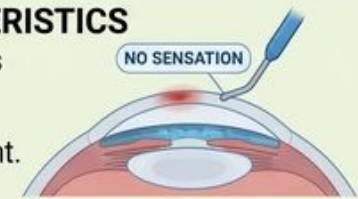
CLINICAL PICTURE (SYMPTOMS & SIGNS)

Marked ocular pain, photophobia, lacrimation, loss of corneal luster. **Diagnostic paracentral dendritic shape with crenated edges** (irregular linear, side branches, knobs).



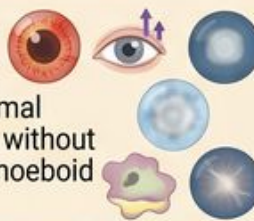
KEY CHARACTERISTICS

Strictly superficial, never perforates (unless complicated).
Profound corneal hypesthesia (numbness) due to nerve involvement.
Chronic, highly recurrent.



COMPLICATIONS

Herpetic iridocyclitis, secondary glaucoma, **Disciform keratitis** (central gray discoid stromal opacity), **Keratitis meta-herpetica** (recurrent without active virus), secondary bacterial infection (amoeboid ulcer with hypopyon), permanent nebulae.



TREATMENT

Local cycloplegics (**atropine**), broad-spectrum antibiotics, eye bandage.
General support (Vitamins A, C).
AVOID HOT FOMENTATION.
Specific antiviral: **Topical Acyclovir (Zovirax) ointment**, q4h, continue 4 days after epithelization.



HERPES ZOSTER OPHTHALMICUS (HZO) & FUNGAL CORNEAL ULCERS

Herpes Zoster Ophthalmicus (HZO)

Definition: A primary infective neuro-dermatitis and keratitis caused by reactivation of the herpes zoster virus within the Gasserian ganglion, tracking down the ophthalmic division of the Trigeminal nerve.



Clinical Picture:

An incubation period of 12–14 days is followed by fever, malaise, and severe neuralgic pain. It presents with a strictly unilateral distribution of painful skin vesicles on an erythematous base following the frontal branch (most common), lacrimal branch, or nasociliary branch.



Hutchinson's Sign: Vesicles on the tip and side of the nose (nasociliary involvement) serve as a strong clinical predictor of severe intraocular inflammation.



Ocular Manifestations:

- Punctate epithelial keratitis (staining selectively with Rose Bengal dye), superficial epithelial vesicles that ulcerate (non-dendritic but fluorescein-positive), subepithelial/disciform keratitis, corneal hypesthesia leading to neurotrophic keratitis,
- scleritis, toxic iridocyclitis with secondary glaucoma,
- extraocular muscle palsies (3rd, 4th, or 6th nerve involvement),
- optic neuritis, and lagophthalmos/ectropion from facial nerve palsy.



Treatment: Rest, sedation, analgesics, calamine lotion/antibiotic powder for eruptive skin lesions, and local steroid skin ointments. Specific treatment requires oral Acyclovir (Zovirax) high-dose tablets for 10 days, while intraocular inflammation (keratitis/iridocyclitis) is managed with local steroid drops and atropine. Residual opacities are managed via keratoplasty.



Fungal Corneal Ulcers

Types: Yeast fungi (*Candida* species) and Filamentous fungi (*Fusarium* and *Aspergillus* species).



Candida Ulcers:

Typically occur in compromised eyes (prolonged steroid use, severe dry eye, exposure keratitis, or HSV keratitis); they present as dense suppurative stromal infiltrates with a distinct oval outline, a plaque-like surface, and surrounding satellite lesions.



Filamentous Fungal Ulcers:

Typically infect healthy corneas following mild trauma involving vegetable or agricultural matter. They present with a characteristic gray or dirty-white dry textured surface, elevated margins, feathery extensions tracking beneath the epithelium into the stroma, multiple micro-abscess satellite lesions, an endothelial plaque, and a severe anterior chamber reaction with hypopyon.



Diagnosis:



Corneal scraping for Gram, Giemsa, and calcofluor white stains, cultured on Sabouraud's dextrose agar and brain-heart infusion broth. Deep lesions may necessitate a superficial keratectomy corneal biopsy.



Sabouraud BHI broth

Treatment:



Mechanical epithelial debridement, topical/systemic antifungal agents (e.g., Natamycin, Amphotericin B, Voriconazole), and local atropine. Resistant cases require a conjunctival flap or therapeutic keratoplasty.



Primary Non-Infective Ulcers: Neurotrophic & Lagophthalmos Keratitis

Neurotrophic Keratitis

Definition & Cause:

A primary non-infective keratitis resulting from the **complete absence of corneal sensation** caused by paralysis of the ophthalmic division of the Trigeminal nerve.



Clinical Features:

Presents with central defective vision and ciliary injection, but features a **complete and characteristic absence of pain**.

Manifests as a central, perfectly disc-shaped neurotrophic ulcer with punched-out edges that **shows zero spontaneous tendency to heal**, progressing rapidly in depth until it perforates.



Treatment:

Focused on maintaining moisture and mechanical protection via **artificial tear drops** by day, lubricating ointment at bedtime, **punctal occlusion**, therapeutic **soft contact lenses**, and surgical **median or paramedian tarsorrhaphy**.



Lagophthalmos Keratitis (Exposure Keratopathy)

Definition & Cause:

A primary non-infective exposure keratitis resulting from **incomplete closure of the palpebral fissure** (voluntary or involuntary), leading to improper wetting and subsequent desiccation and ulceration of the corneal surface.



Etiology & Clinical Picture:

Caused by severe ectropion, facial nerve palsy, lid coloboma, overcorrected ptosis surgery, thyroid orbitopathy, proptosis, buphthalmos, high debility, or general anesthesia.

Stitching pain, reflex lacrimation, ciliary injection, severe dry eye signs, and a toxic iritis with miosis and flare. The characteristic **exposure ulcer occupies the lower third of the cornea**, demonstrating a **sharp, straight upper horizontal boundary line**.



Treatment:

Protective **eye bandaging**, treating the primary cause of lagophthalmos, utilizing hourly **artificial tears**, therapeutic **contact lenses**, and performing an **urgent tarsorrhaphy** to safeguard the globe from perforation.



KERATOCONUS (Conical Cornea) & ECTATIC OCULAR DISORDERS



DEFINITION

A bilateral, slowly progressive developmental weakness of the central part of the corneal stroma. Under the thinning effect of normal intraocular pressure, it bulges forward to assume a distinct conical projection.



CLINICAL FEATURES & DEMOGRAPHICS

Typically manifests at puberty and is more prevalent in females, frequently demonstrating a positive family history. It runs a gradual onset, slowly progressive course that tends to stabilize after several years. Rarely, it features an acute onset or relapse with sudden breaks in Descemet's membrane, allowing massive aqueous influx and severe corneal edema, known as ACUTE HYDROPS.



SYMPTOMS

Gradual, progressive, painless diminution of vision caused by axial and curvature myopia, severe irregular astigmatism, and apical corneal opacities.



CLINICAL SIGNS (EARLY CASES)

Marked central corneal thinning and progressive ectasia; increased corneal power and steep curvature on Keratometry; distortion of the mires on Placido disc examination; and a spinning scissor-like red reflex on Retinoscopy.

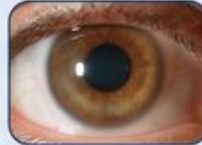


CLINICAL SIGNS (ADVANCED CASES)



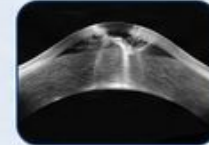
MUNSON'S SIGN

A characteristic angulation or V-shaped indentation of the lower eyelid margin when the patient looks downward.



FLEISCHER'S RING

A partial or complete brown ring at the base of the cone, caused by hemosiderin iron deposition within the basal epithelial cells.



APICAL SIGNS

Folds in the deep stroma, ruptures in Bowman's membrane filled with fibrous tissue, and apical opacities at the tip of the cone, which is usually directed inferiorly and nasally.



DIFFERENTIAL DIAGNOSIS

Differentiated from other corneal ectatic states including KERATOECTASIA (bulging of a scarred cornea), ANTERIOR STAPHYLOMA (ectatic scar lined by uveal tissue), and KERATOglobus (generalized hemispherical thinning and protrusion of the entire cornea).



MANAGEMENT PROTOCOL



General support via a nutritious diet and vitamins. Local management consists of spectacles in very early cases; RIGID GAS PERMEABLE (RGP) hard contact lenses to correct irregular astigmatism by neutralizing corneal power; CORNEAL COLLAGEN CROSS-LINKING (CXL) using Riboflavin and UV-A light to arrest progression; INTRASTROMAL CORNEAL RING SEGMENTS (ICRS) if the patient is intolerant to contact lenses; and PENETRATING KERATOPLASTY if lenses fail or deep apical opacities compromise vision.

CORNEAL PROCEDURES: Keratoplasty & Keratorefractive Surgery

Keratoplasty (Corneal Graft)

Definition:

The surgical excision of a diseased, opaque, or structurally compromised section of the host cornea and its subsequent replacement with a clear donor corneal graft.

Primary Surgical Types:

- 1) **Lamellar Keratoplasty:** Transplantation of only the superficial layers (half-thickness) of the cornea, indicated for strictly superficial opacities.
- 2) **Penetrating Keratoplasty:** Full-thickness transplantation involving all layers of the cornea.

Clinical Indications:

- Optical (to restore visual acuity in dense corneal opacities and advanced keratoconus);
- Therapeutic (to eradicate resistant active disease like chronic dendritic ulcers or secure a corneal fistula);
- Structural/Tectonic (to restore globe integrity following excision of a massive pterygium or symblepharon).

Absolute Contraindications:

- Untreated or uncontrolled glaucoma, active anterior uveitis, severe xerosis (dry eye), neuromyopathic keratitis, and deep stromal corneal neovascularization.

Donor Graft Criteria:

- Must be harvested from tissues free of local ocular pathology and completely clear of transmissible infectious diseases including HIV and hepatitis viruses.

Complications:

- Operative (lens injury, vitreous loss, expulsive choroidal hemorrhage);
- Postoperative (microbial infection, glaucoma, astigmatism, and graft rejection).



Keratorefractive Surgery

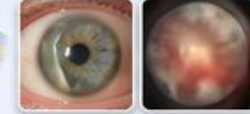
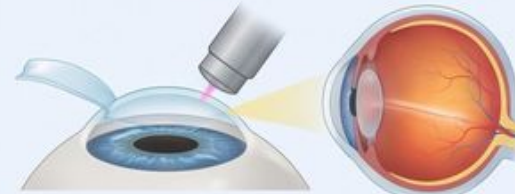
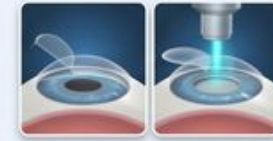
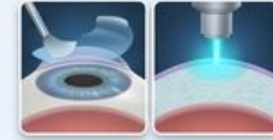
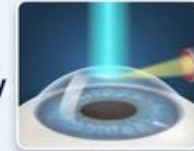
Definition:

Specialized corneal surgical procedures designed to modify the anterior curvature and refractive power of the cornea to permanently correct ametropic errors of refraction.

Primary Laser Modalities:

- 1) **Photorefractive Keratectomy (PRK): Surface ablation** where the Excimer laser is applied directly to remove a precise amount of the anterior corneal stroma after epithelial removal, modifying the surface curvature.

- 2) **Laser In Situ Keratomileusis (LASIK):** Excimer reveals the endothelium. Involves creating a superficial corneal flap of 130–160 micrometers thickness using a microkeratome or femtosecond laser. The flap is reflected, and the Excimer laser is applied to sculpt the deep stromal bed (flattening the cornea to correct myopia), after which the flap is carefully repositioned without sutures.



CHAPTER 05

Chapter 5: Lens

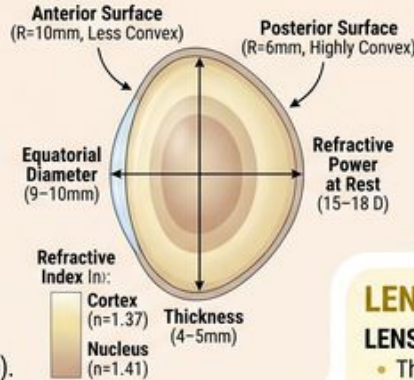


APPLIED ANATOMY, HISTOLOGY, & PHYSIOLOGY OF THE CRYSTALLINE LENS • CONGENITAL AND DEVELOPMENTAL CATARACT: ETIOLOGY, MORPHOLOGY, AND COMPLICATIONS • ACQUIRED SENILE CORTICAL CATARACT: STAGES AND MORPHOLOGY • SENILE NUCLEAR CATARACT AND NUCLEAR SCLEROSIS DIFFERENTIAL DIAGNOSIS • LENS-INDUCED GLAUCOMAS AND LENS DISPLACEMENTS (ECTOPIA LENTIS) • PREOPERATIVE OCULAR ASSESSMENT, BIOMETRY, AND INTRAOCULAR LENS (IOL) MODALITIES • MODERN CATARACT SURGICAL TECHNIQUES AND COMPLICATIONS • TRAUMATIC, COMPLICATED, AND AFTER-CATARACT (POSTERIOR CAPSULE OPACIFICATION)

APPLIED ANATOMY, HISTOLOGY, & PHYSIOLOGY OF THE CRYSTALLINE LENS

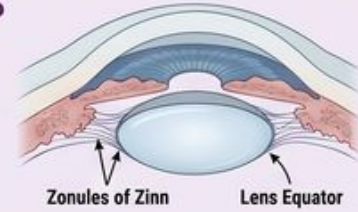
APPLIED ANATOMY & DIMENSIONS

- The crystalline lens is a **biconvex**, **transparent**, **avascular**, and **elastic** structure.
- The **anterior surface** is less convex (radius of curvature 10 mm), while the **posterior surface** is highly convex (radius of curvature 6 mm). They meet at the **equator**.
- It has an equatorial **diameter** of 9–10 mm, a **thickness** of 4–5 mm, and a **refractive power** of 15–18 Diopters at rest, which **increases significantly during accommodation**.
- The **total refractive index** is 1.39 (ranging from 1.37 in the cortex to 1.41 in the dense core nucleus).



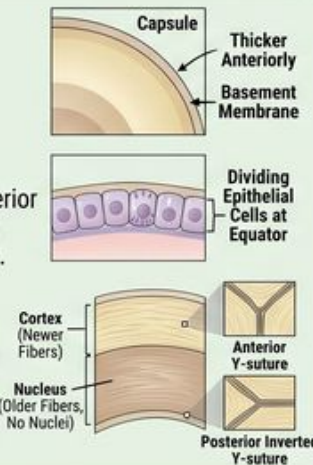
SUSPENSORY LIGAMENTS (ZONULES OF ZINN)

- Fibrillar structures** arising from the ciliary body epithelium and inserting into the lens equator to **hold it in position**.



HISTOLOGICAL STRUCTURE

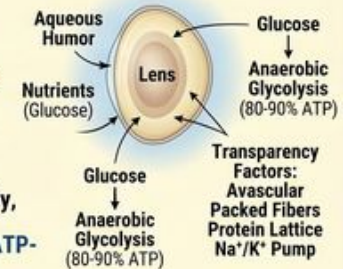
- Capsule:** A **smooth, elastic, semipermeable basement membrane** secreted by the epithelium, **thicker anteriorly** than posteriorly.
- Anterior Epithelium:** A single layer of **cuboidal metabolic cells** located exclusively beneath the anterior and equatorial capsule. They **continuously divide at the equator** to form new lens fibers throughout life.
- Lens Fibers:** **Elongated, tightly packed cells** that **lose their nuclei** as they age, forming the **cortex** (newer outer fibers) and the **nucleus** (older compressed central fibers). They meet at the **anterior Y-shaped** and **posterior inverted Y-shaped sutures**.



LENS METABOLISM, TRANSPARENCY, & ACCOMMODATION

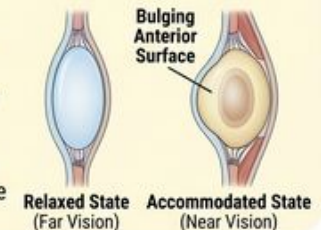
LENS METABOLISM & TRANSPARENCY

- The lens is completely **avascular** and relies on the **aqueous humor** for nutrition.
- It derives **80–90% of its energy (ATP)** via **anaerobic glycolysis (Embden-Meyerhof pathway)**, with the remaining via the HMP shunt.
- In diabetic states, the **Sorbitol pathway** becomes abnormally active.
- Transparency is maintained by **absolute avascularity**, **tightly packed fiber membranes**, a **regular lattice arrangement** of crystalline proteins, and an **active ATP-dependent sodium-potassium dehydration pump** located in the epithelium to prevent fluid accumulation.



MECHANISM OF ACCOMMODATION

- The process of **increasing lens dioptric power to focus near objects**.
- Contraction of the ciliary muscle** moves the ciliary body forward and inward, **relaxing the zonules of Zinn**.
- Due to its **intrinsic elasticity**, the lens capsule **bulges forward**, making the **anterior surface highly convex** and increasing its refractive index, while the posterior surface remains almost unchanged.



CONGENITAL AND DEVELOPMENTAL CATARACT: ETIOLOGY, MORPHOLOGY, AND COMPLICATIONS

• **Definition:** An opacity of the crystalline lens or its capsule present at birth or developing during early infancy. 



COMPREHENSIVE ETIOLOGY



1) **Hereditary: Autosomal dominant inheritance** is the most common genetic cause.



2) **Maternal Intrauterine Infections (TORCH):** Occurring primarily during the first trimester when the lens vesicle detaches. Includes **Rubella virus, Toxoplasmosis, Cytomegalovirus, and Herpes simplex.**



3) **Maternal Nutritional & Systemic Factors**
Severe malnutrition, avitaminosis (Vitamin D and Calcium deficiencies), and systemic hypoxia or placental hemorrhages.



4) **Infantile Metabolic Disorders**
Galactosemia (classic galactose-1-phosphate uridylyltransferase deficiency, presenting with a characteristic "oil-droplet" central lens opacity), **Galactokinase deficiency, Neonatal Hypoglycemia, Lowe (Oculo-cerebro-renal) syndrome.**



5) **Chromosomal Anomalies**
Down syndrome (Trisomy 21), Edwards syndrome, and Patau syndrome.



MORPHOLOGICAL CLASSIFICATION



1) Anterior Polar Cataract

A small opacity at the anterior pole, often caused by delayed anterior chamber formation. Usually stationary and visually non-significant.



2) Posterior Polar Cataract

Sits directly at the posterior pole near the nodal optical point. Causes a **severe drop in vision** and is associated with persistent hyperplastic primary vitreous (PHPV).



3) Zonular / Lamellar Cataract

The **commonest symptomatic** developmental type. Affects a specific layer or zone of the lens cortex while the nucleus and outer cortex remain clear. Shows central disc opacification with linear "riders" or spokes projecting outward like a **steering wheel**.



4) Blue Dotted (Cataracta Punctata)

Tiny, bluish, round dots scattered throughout the cortex; strictly stationary and asymptomatic.



5) Coronary Cataract

Manifests around puberty as a ring or corona of **peripheral club-shaped opacities**; leaves the **visual axis free**.



6) Rubella Total Cataract

Dense, pearly white total opacification at birth. Associated with **microphthalmos, pigmentary retinopathy**, and systemic defects (deafness, patent ductus arteriosus, and mental retardation).



COMPLICATIONS & SEQUELAE

- Visually significant central cataracts obstruct light projection, leading to **irreversible sensory deprivation amblyopia (lazy eye)**. Unilateral cases cause **sensory strabismus (squint)**, and bilateral cases lead to **generalized horizontal nystagmus** and **delayed mental development**.



CHOICE OF TREATMENT

- Visually non-significant opacities require conservative observation and refraction tracking. Visually significant dense opacities require **urgent surgical extraction via Lensectomy and Anterior Vitrectomy** within the **first 6-8 weeks of life** to safeguard macular development, followed by visual rehabilitation using **contact lenses, pediatric spectacles, or primary intraocular lens (IOL) implantation** if the child is over over 2 years old.



ACQUIRED SENILE CORTICAL CATARACT: STAGES & MORPHOLOGY

DEFINITION & PATHOGENESIS

An **acquired, progressive opacification**

An **acquired, progressive opacification** of the lens cortex, occurring in individuals **>50 years** due to **AGE-RELATED OXIDATIVE STRESS**



These **two major changes** *alavoxly* impress healthy opacorratioⁿ ao ner level and maio proposed. invenrement, lens cortex, done o^o compury prograssies and a capsulmy lens.

It is structurally **BILATERAL** but clinically **ASYMMETRICAL**



Bilateral, Asymmetrical

FOUR SUCCESSIVE CLINICAL STAGES

1) INCIPIENT STAGE

emphas early fluid accumulation

WATER CLEFTS or **LAMELLAR SEPARATION**



Cuneiform Opacities

- mild progressive diminution of vision
- monocular diplopia/polyopia
- halos around lights

Visual acuity normal or slightly reduced Halos



2) IMMATURE STAGE

opacification extends deeper and **INCOMPLETE**
Symptoms: **PROGRESSIVE DAY & NIGHT BLINDNESS**
significantly dropped, better than hand movement



UNIFORM GRAYISH-WHITE LENS

POSITIVE IRIS SHADOW TEST
(clear cortex layer exists)

*INTUMESCENT FORM (CRITICAL VARIANT)

emphasized rapid fiber breakdown and **MASSIVE OSMOTIC FLUID INFLUX**



Lens Swells, Pushes Iris Forward

ACUTE SECONDARY PHACO-MORPHIC ANGLE-CLOSURE GLAUCOMA

3) MATURE STAGE

emphasized **TOTAL OPACIFICATION**
visual acuity **DROPS SEVERELY**
HAND MOVEMENT / FINGER COUNTING



COMPLETELY OPAQUE, PEARLY-WHITE/CHALKY-WHITE LENS
IRIS SHADOW COMPLETELY NEGATIVE
(No clear cortex remains)

4) HYPERMATURE STAGE

enzymatic disintegration & **LIQUEFACTION**, water escapes, lens to **SHRINK**

• SCLEROTIC / SHRUNKEN FORM

emphasized small, flat, yellow disc
DEEPLY WRINKLED ANTERIOR CAPSULE
calcium/cholesterol deposits
DEEP ANTERIOR CHAMBER
TREMULOUS IRIS (IRIDODONESIS)



• MORGAGNIAN FORM

ENTIRE CORTIX LIQUEFIES
MILKY-WHITE FLUID
DENSE, HARD, BROWNISH-YELLOW NUCLEUS SINKS
IRIS SHADOW REAPPEARS



SENILE NUCLEAR CATARACT & NUCLEAR SCLEROSIS DIFFERENTIAL DIAGNOSIS

SENILE NUCLEAR CATARACT: Definition, Symptoms & Signs

- Definition & Pathology:** An acquired **opacification** of the central lens core occurring after the age of 50. The primary mechanism is an **exaggerated, progressive physical compaction and dehydration** of the central nuclear fibers, accompanied by **protein denaturation and accumulation** of urochrome and melanin pigments, turning the nucleus **yellow, brown (cataracta brunescens), or black (cataracta nigra)**.



- Clinical Symptoms:**



Day Blindness
(Bright Light)

"Second Sight"
(Near Vision,
Temporary)

- Clinical Signs:**



Slit-lamp: Dark
Central Core

Red Reflex:
Central Black Disc

Gradual, painless progressive diminution of vision. In the early stages, the marked increase in the refractive index of the sclerosed nucleus induces a **progressive index myopia**. This index change shifts the patient's near point closer, temporarily correcting presbyopia and allowing elderly patients to read again without reading glasses—a clinical phenomenon termed "**second sight**." As the central opacity darkens, it causes **day blindness** (vision worsens in bright light due to pupillary constriction blocking the central visual axis).

Visual acuity progressively drops down to hand movement. **Slit-lamp evaluation** reveals a distinct **yellow, brown, or black central core**. The peripheral cortex remains long clear; thus, the **iris shadow test remains positive** for an extended duration. On distant direct ophthalmoscopy, the **red reflex test** reveals a dense **black central disc** against a bright red peripheral ring.

DIFFERENTIAL DIAGNOSIS: Nuclear Sclerosis vs. Pathological Nuclear Cataract



Physiological Nuclear Sclerosis

Represents a **normal age-related physiological hardening** and compaction of lens fibers.

Visual acuity is easily **corrected back to normal (6/6)** using appropriate myopic spectacles.

On oblique illumination, the lens displays a **faint grayish sheen**, distant direct ophthalmoscopy shows a **completely clear, unobstructed red reflex**.

Treatment relies solely on **updating spectacle prescriptions**.



Pathological Nuclear Cataract

Represents a **true pathological opacification and discoloration**.

Visual acuity **cannot be improved or corrected** with spectacles.

Oblique illumination reveals a **dense yellow, dark brown, or black central core**, and the red reflex demonstrates a **permanent central black silhouette defect**.

Treatment requires **surgical extraction** once it interferes with daily activities.



LENS-INDUCED GLAUCOMAS & LENS DISPLACEMENTS (ECTOPIA LENTIS): Explanatory Graphic

Pathologies and Mechanisms Impacting Intraocular Pressure and Lens Position

LENS-INDUCED (PHACOGENIC) GLAUCOMAS

Lens-Induced (Phacogenic) Glaucomas: Secondary Hypertensive Conditions Directly Caused by Lens Pathology

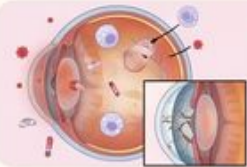


1) Phacomorphic Glaucoma

Secondary **angle-closure glaucoma** occurring during the **intumescent stage**. The swollen, **hydrated lens** pushes the **iris-lens diaphragm forward**, causing **pupillary block** and mechanical closure of the **anterior chamber angle**.

2) Phacolytic Glaucoma

Secondary **open-angle glaucoma** occurring in **hypermaturing cataracts**. Soluble **lens proteins leak** through the intact but degenerated capsule into the **anterior chamber**. **Macrophages phagocytose** these heavy proteins and **clog the trabecular meshwork**, causing a **rapid rise in intraocular pressure**. Presents with a **deep anterior chamber**, **floating white particles**, and **severe pain**.

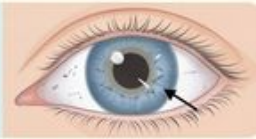


3) Phacoanaphylactic Endophthalmitis

An **autoimmune, granulomatous intraocular inflammation** triggered by exposure to **lens proteins** following **traumatic or surgical rupture of the lens capsule**, leading to secondary **open or closed-angle glaucoma**.

4) Lens-Particle Glaucoma

Occurs when **free cortical remnants block the trabecular meshwork** after **trauma or incomplete cataract extraction**.

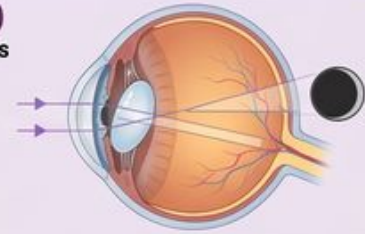


LENS DISPLACEMENTS (ECTOPIA LENTIS)

Lens Displacements (Ectopia Lentis): Disruption of the Zonules of Zinn Leading to Malposition of the Lens

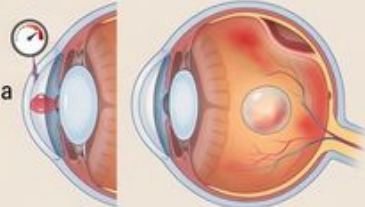
1) Subluxation (Partial Displacement)

Some **zonular fibers are broken**, and the **lens tilts** but remains **behind the pupil**. Signs include a **shallow or irregular anterior chamber depth**, a **tremulous iris (iridodonesis)**, a **tremulous lens (phacodonesis)**, **monocular diplopia** (as light passes through both the aphakic and phakic zones), and the **crescentic edge of the lens** becomes visible upon **full pupillary dilation**.



2) Dislocation (Complete Displacement)

All zonular fibers are ruptured, and the lens is **completely displaced** from the pupillary area. It can dislocate **anteriorly into the anterior chamber** (causing acute pupillary block glaucoma and corneal endothelial damage) or **posteriorly into the vitreous cavity** (causing **vitreous liquefaction, uveitis, or retinal detachment**, leaving the **pupillary area completely aphakic**).



Primary Causes of Ectopia Lentis



Trauma (most common cause of unilateral displacement) and **Congenital/Hereditary conditions** (usually bilateral, including **Marfan syndrome** where the lens displaces superotemporally, **Homocystinuria** inferomedially with a high risk of thromboembolism, and **Weill-Marchesani syndrome** causing microspherophakia and anterior dislocation).



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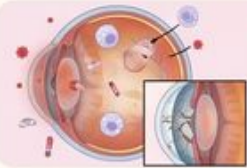


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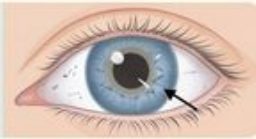


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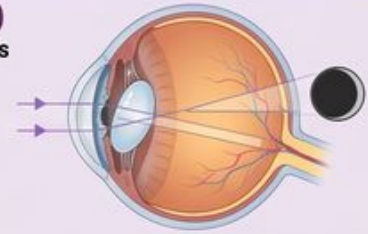


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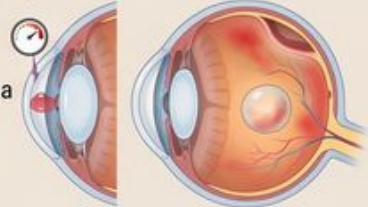
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MODERN CATARACT SURGICAL TECHNIQUES & COMPLICATIONS: A COMPREHENSIVE OVERVIEW

I. TIMELINES FOR INTERVENTION

- OPERATED WHEN VISUAL IMPAIRMENT interferes with professional or daily activities, OR regardless of vision if LENS-INDUCED COMPLICATIONS occur (intumescence, hypermaturity, or phacolytic glaucoma).



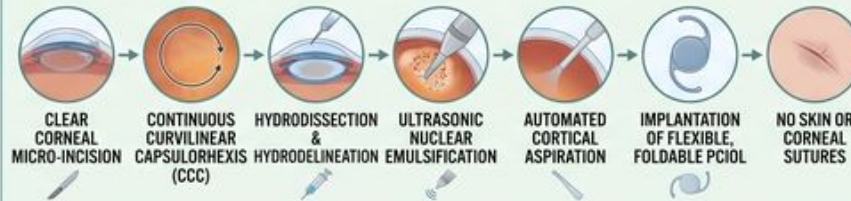
- SIMULTANEOUS BILATERAL SURGERY IS CONTRAINDICATED to avoid bilateral endophthalmitis; a 1-WEEK INTERVAL IS MAINTAINED.



II. PRIMARY SURGICAL TECHNIQUES

1) PHACOEMULSIFICATION: THE MODERN GOLD STANDARD

Performed under topical anesthesia through a self-sealing micro-incision (2.2–2.8 mm).



2) MANUAL EXTRACAPSULAR CATARACT EXTRACTION (ECCE)

Indicated for extremely hard, advanced cataracts where phacoemulsification might damage corneal endothelium.



3) MANUAL SMALL INCISION CATARACT SURGERY (MSICS)

A hybrid manual ECCE technique using a self-sealing SCLERO-CORNEAL TUNNEL INCISION that requires NO SUTURES.



4) INTRACAPSULAR CATARACT EXTRACTION (ICCE)

COMPLETE REMOVAL of the entire lens within its intact capsule through a MASSIVE 12 mm INCISION using a cryoprobe. OBSOLETE NOWADAYS due to high vitreous loss, aphakia, and retinal detachment, but indicated for severe lens subluxation/dislocation.



III. MAJOR POSTOPERATIVE COMPLICATIONS

OPERATIVE VITREOUS LOSS (Posterior capsule rupture icon, vitreous leaking)

EXPULSIVE CHOROIDAL HEMORRHAGE

POSTOPERATIVE TRANSIENT CORNEAL EDEMA (Striate keratopathy, cloudy cornea with lines icon)

SECONDARY GLAUCOMA

CYSTOID MACULAR EDEMA (Irvine-Gass syndrome, macular area icon with cysts)

RETINAL DETACHMENT

DEVASTATING ACUTE BACTERIAL ENDOPHTHALMITIS (Severe infection, presents with SEVERE PAIN, LOSS OF VISION, HYPOPYON, and VITREOUS HAZE within 48–72 hours; REQUIRES URGENT INTRAVITREAL ANTIBIOTIC INJECTIONS)

TYPES & PATHOLOGY OF CATARACTS

• Traumatic Cataract: Opacification due to Ocular Trauma



1) **Concussion (Blunt) Trauma:** Causes a characteristic "Rosette-shaped" or feathery cortical opacity, typically tracking along posterior lens sutures from aqueous forced into lamellae. May show a "Vossius ring" (brown iris pigment imprint on anterior capsule).



2) **Perforating (Sharp) Trauma:** Direct lens capsule rupture leads to rapid, massive hydration, presenting as a soft, white intumescent cataract. Can rapidly swell and leak cortical remnants into the AC.



3) **Physical/Radiation Injuries:** Infrared radiation causes "Glass Blower's Cataract" (true anterior capsule exfoliation). Ionizing radiation (X-rays), and Electrical burns produce anterior subcapsular punctate opacities.



• Secondary (Presenile) Cataract: Bilateral, Early-Onset due to Systemic Metabolic Diseases



• **Diabetes Mellitus:** Presents with classic 'snowflake' subcapsular opacities and rapid progression.



• **Hypoparathyroidism:** Causes tetany cataract.



• **Myotonic Dystrophy:** Shows fine polychromatic cortical crystals.

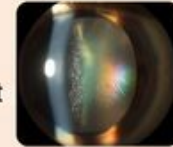


• Complicated Cataract: Secondary to Severe, Chronic Intraocular Diseases



Intraocular Causes: Chronic anterior uveitis/iridocyclitis, heterochromic iridocyclitis, high axial myopia, advanced retinitis pigmentosa, long-standing retinal detachment, intraocular tumors.

Pathogenesis & Morphology: Diffusing toxins or metabolic imbalances alter lens permeability. Characteristically manifests as a bread-crumble-like posterior subcapsular cortical opacity displaying a distinct irregular "polychromatic luster" (iridescent gold, green, and blue sheen) under slit-lamp evaluation.



• After-Cataract (Posterior Capsule Opacification / PCO):



Definition: Opacification and thickening of the remaining posterior lens capsule persisting or developing months to years after extracapsular cataract extraction or phacoemulsification.

Pathophysiology & Signs: Proliferation and migration of residual equatorial lens epithelial cells. Can cluster into transparent, soap-bubble-like 'Elschnig's pearls', or cause dense capsular fibrosis forming 'Soemmering's ring'. Presents with a gradual, painless drop in visual acuity.



Management: Non-invasive, outpatient 'Neodymium:YAG (Nd:YAG) laser capsulotomy' to create a clear central opening in the pupillary visual axis, instantly restoring vision.



CHAPTER 06

Chapter 6: Orbit



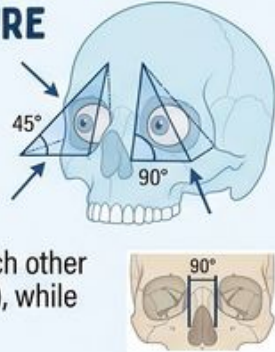
ANATOMY OF THE ORBIT (WALLS, FORAMINA, SINUSES, AND CONTENTS) • ORBITAL INFECTIONS: ORBITAL CELLULITIS AND PRESEPTAL CELLULITIS • THYROID EYE DISEASE (TED) & THYROTOXIC EXOPHTHALMOS (GRAVES' DISEASE) • CAVERNOUS SINUS THROMBOSIS (VASCULAR ORBITAL LESION) • PROPTOSIS, ENOPHTHALMOS, AND ATROPHIA BULBI • MAJOR OPERATIONS OF THE ORBIT (ENUCLEATION, EVISCERATION, AND EXENTERATION)

ANATOMY OF THE ORBIT: WALLS, FORAMINA, SINUSES, AND CONTENTS

• DEFINITION & STRUCTURE

The **orbit** is a **four-sided pyramidal space** on both sides of the midline containing the globe.

The adult lateral orbital walls are approximately **90 degrees** from each other (45 degrees from the sagittal plane), while the medial walls are **parallel**.



• FORAMINA & TRANSMISSIONS

Optic Foramen

Transmits the **optic nerve** (with meninges), **ophthalmic artery**, and sympathetic fibers



Superior Orbital Fissure

Transmits **oculomotor** (3rd), **trochlear** (4th), and **abducent** (6th) nerves, the **three branches of the ophthalmic division of the trigeminal nerve**, superior and inferior ophthalmic veins, and sympathetic fibers



Inferior Orbital Fissure

Transmits the **maxillary division of the trigeminal nerve** and **infra-orbital artery**



• WALLS OF THE ORBIT

- Comprised of **seven bones**: **ethmoid, frontal, lacrimal, maxillary, palatine, sphenoid, and zygomatic**.

• Roof

Mainly formed by the **orbital plate of the frontal bone**



• Floor

Composed of the **maxillary, palatine, and zygomatic bones**



Medial Wall

The **thinnest** the **ethmoid bone** and **maxilla**, mainly view of medial bones



• Medial Wall

The **thinnest** (0.2 to 0.4 mm) and **smallest**, mainly made of the **ethmoid bone** with contributions from the maxillary and lacrimal bones

Lateral Wall

Bounded by the **superior and inferior orbital fissures**



• INTRA-ORBITAL CONTENTS & SINUSES

Intra-orbital Contents: The **eyeball**, **extraocular muscles**, **lacrimal gland**, **vessels and nerves**, **ciliary ganglion**, and **orbital fat**.



The surrounding bony plates are **intimately pneumatized by the paranasal sinuses**.



ORBITAL CELLULITIS (Postseptal Infection)

Figure 3: Orbital cellulitis (Image of an eye with severe eyelid swelling, redness, and proptosis)



❖ ETIOLOGY

✓ SOURCES OF INFECTION:

- 1- **Endogenous**: extension from neighboring structures as the **paranasal sinuses**
- 2- **Exogenous**: as in penetrating trauma or septic orbital surgery
- 3- **Blood born**: metastatic infection as in **pyemia**

✓ CAUSATIVE ORGANISMS: pyogenic organisms specially **staphylococci**

❖ CLINICAL PICTURE

✓ SYMPTOMS:

- A. General: Fever, malaise, headache...etc B. Local: 1. severe orbital pain
2. Diplopia

✓ SIGNS:

Lid edema, redness and tenderness



Conjunctival chemosis
Axial proptosis



Limitation of ocular movements in all directions



❖ COMPLICATIONS

A- Ocular:

Exposure keratopathy



Orbital Abscess & Panophthalmitis



Central retinal vein or artery occlusion



N.B: In severe untreated cases, **systemic complications** may develop, e.g. **septicemia** and **pyemia**

B- Intracranial: Meningitis



Brain abscess



Cavernous sinus thrombosis



❖ DIFFERENTIAL DIAGNOSIS

Cavernous sinus thrombosis



Acute suppurative dacryoadenitis



Preseptal cellulitis



❖ TREATMENT

- **Hospitalization** is necessary in this situation for **close follow-up**
- **Topical** and **systemic parenteral broad-spectrum antibiotics**
- **Hot fomentations**

N.B: If abscess is formed, it should be **drained**

PRESEPTAL CELLULITIS (Periorbital Infection)

(Image of an eye with swelling limited to the eyelid, no proptosis)



❖ DEFINITION & FREQUENCY

- **Definition**: infection of the eyelid and surrounding skin **anterior to the orbital septum**
- Preseptal cellulitis is **far more common** than orbital cellulitis



❖ ETIOLOGY

- Preseptal cellulitis is usually caused by **contiguous spread of infection** from local **facial or eyelid injuries**, **insect or animal bites**, **conjunctivitis**, **chalazion**, or **sinusitis**



❖ CLINICAL PICTURE

✓ SYMPTOMS: Swelling, pain, redness of the eyelid and sometimes fever

✓ SIGNS:

Tenderness, swelling, warmth, redness or discoloration of the eyelid

- Patients may be **unable to open their eyes** because of eyelid swelling
- The swelling and discomfort can make it difficult to examine the eye, but when done, examination shows that **visual acuity is NOT affected**, **ocular movement is intact**, and the **globe is NOT pushed forward (proptosis)**

E
T
P
20/20



NO Proptosis

This is very important to differentiate it from orbital cellulitis

❖ TREATMENT

- Treatment is the **same as orbital cellulitis**



❖ N.B. DIAGNOSIS

- **N.B:** Diagnosis of preseptal cellulitis and orbital cellulitis is **mainly clinical**
- However, if the **diagnosis is difficult**, **CT or MRI orbit** can be done
- They **differentiate easily** between the two conditions
- Soft tissue involvement is present **anterior to the orbital septum** in preseptal cellulitis while it is present **posterior to the orbital septum** in case of orbital cellulitis

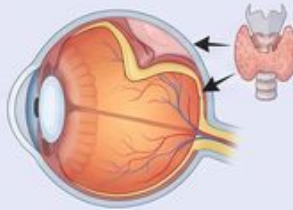


Thyroid Eye Disease (TED) & Thyrotoxic Exophthalmos (Graves' Disease)

DEFINITION

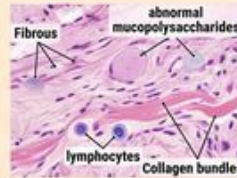
Ocular and orbital changes resulting from **thyroid gland dysfunction**.

It is the **most common cause** of both unilateral and bilateral **proptosis**, and can occur in **hyperthyroid, hypothyroid, or euthyroid** patients.



PATHOGENESIS

Immunogenic deposition of **abnormal mucopolysaccharides and collagen** in the orbit and extraocular muscles, accompanied by **lymphocytic infiltration**. This swelling is followed later by **permanent muscle fibrosis**.



THYROTOXIC EXOPHTHALMOS (GRAVES' DISEASE)

More common in **middle-aged females** with **hyperthyroidism**.



TREATMENT

- Treat the **underlying thyrotoxicosis**, use **artificial tear substitutes** for corneal protection, perform **lateral tarsorrhaphy** for severe lagophthalmos to prevent **exposure keratopathy**, administer **systemic steroids**, and perform **surgical orbital decompression** in advanced cases with **optic neuropathy**.



CLINICAL PICTURE

General Symptoms:

- Nervousness, irritability, heat intolerance, palpitation, weight loss, enlarged thyroid, rapid pulse, and tremors.**



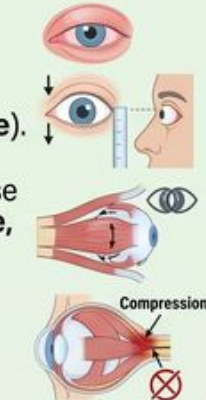
Local Symptoms:

- Ocular discomfort, cosmetic complaints, and diplopia.**



Local Signs:

- Soft tissue involvement** (edema and congestion of lids/conjunctiva).
- Lid retraction** (causes a **staring look, infrequent blinking, and lid lag on down-gaze**).
- Axial proptosis** (unilateral or bilateral).
- Restrictive myopathy** (limited motility because **muscles swell up to 8 times their normal size, causing diplopia**).
- Optic neuropathy** (due to **severe apical compression** by enlarged muscles).



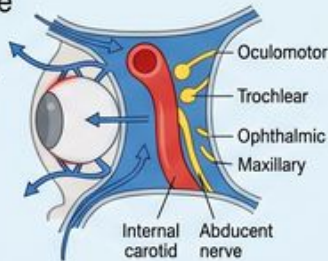
CAVERNOUS SINUS THROMBOSIS (Vascular Orbital Lesion)

ANATOMY

The cavernous sinus is a **wide venous channel** receiving tributaries from the **orbit, face, mouth, pharynx, paranasal sinuses, and middle ear**.

Its lateral wall contains the **oculomotor (3rd)** and **trochlear (4th)** nerves, plus the **ophthalmic and maxillary divisions** of the trigeminal nerve.

The **internal carotid artery** and **abducent (6th) nerve** run directly inside the sinus.



CLINICAL PICTURE

SYMPTOMS

General cerebral symptoms with fever, malaise, and headache.

Local symptoms include **severe orbital pain** and **diplopia** (from oculomotor paralysis).



SIGNS

Lid edema, conjunctival chemosis, rapidly increasing axial proptosis, total ophthalmoplegia, distinct edema in the mastoid region, and rapid involvement of the contralateral eye.



ETIOLOGY

Spread of infection along connected tributaries (often a **fatal complication of orbital cellulitis**) or via **blood-borne infection**.



COMPLICATIONS

Severe meningitis and **brain abscess**.



TREATMENT

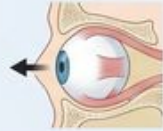
Requires **immediate hospitalization, intensive intravenous broad-spectrum antibiotics, intravenous heparin every 6 hours, and urgent treatment of the causative primary septic focus.**



OCULAR POSITIONAL DISORDERS & END-STAGE CONDITIONS

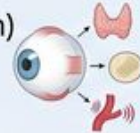
PROPTOSIS (Exophthalmos)

Abnormal protrusion of the eyeball outside the orbital rim.



Unilateral Causes

- **Thyroid ophthalmopathy** (most common)
- developmental dermoid cysts
- trauma (retrobulbar hemorrhage)
- inflammatory orbital pseudotumor
- vascular orbital varices or **carotid-cavernous fistula** (showing pulsating proptosis, thrill, and bruit)
- neoplastic tumors (hemangioma, rhabdomyosarcoma, optic nerve glioma)



Bilateral Causes

- **Thyroid ophthalmopathy** (most common)
- cavernous sinus thrombosis



Clinical Evaluation

Confirm and measure degree of proptosis using a transparent ruler or **Hertel's exophthalmometer** (measuring distance between corneal apex and lateral orbital rim). Note direction: **axial** (intraconal lesions) or **displaced** (extraconal lesions).

Utilize CT, MRI, B-scan ultrasonography, and thyroid profiles.



ENOPHTHALMOS

Retraction of the eyeball **deep into** the orbit.



Causes

- **Microphthalmos**
- orbital floor **blow-out fractures**
- post-inflammatory fibrosis
- cicatrizing orbital tumors
- surgical removal of orbital contents



ATROPHIA BULBI

Completely **shrunk**, disorganized globe.



Etiology

- Severe ocular trauma
- long-standing **retinal detachment**
- severe intraocular **inflammation** (endophthalmitis)



Clinical Picture & Management

Presents as a completely **shrunk, disorganized globe** with **total loss of vision**.

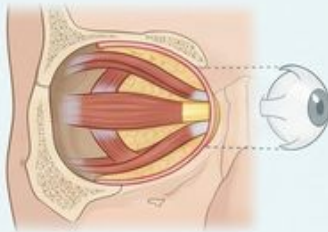
Management requires **surgical enucleation** if it becomes a **blind painful eye**.



MAJOR OPERATIONS OF THE ORBIT: ENUCLEATION, EVISCERATION, & EXENTERATION

ENUCLEATION

Removal of the Eyeball



DEFINITION

Complete removal of the **entire eyeball**, leaving the **extraocular muscles**, the **conjunctiva**, and the **optic nerve** behind in the orbit.

INDICATIONS

- To **stop pain** in blind painful eyes (e.g., *absolute glaucoma*).
- To **save the patient's life** in cases of intraocular tumors (e.g., *retinoblastoma*).
- To **save the contralateral eye** in cases of sympathetic ophthalmia.
- To **improve cosmetic appearance** in severely deformed eyes after trauma using a **prosthesis**.



EVISCERATION

Removal of Eye Contents



DEFINITION

Removal of the **internal intraocular contents only**, leaving the **host scleral shell**, **conjunctiva**, and **optic nerve** fully intact and free.

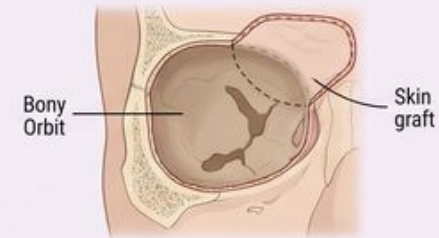
INDICATIONS

- **Active endophthalmitis.**
- **Panophthalmitis.**
- **Uncontrollable expulsive hemorrhage** (where cutting the optic nerve risks spreading infection to the brain).



EXENTERATION

Massive Orbital Removal



DEFINITION

Massive surgical removal of all **orbital contents** completely within the orbital periosteum, **including the eyelids**.

INDICATIONS

- **Severe, destructive malignant orbital tumors** with a free periosteum and no distant metastasis.
- **Massive malignant tumors** of the lid, conjunctiva, and eyeball that **extensively invade the orbit**.



Note: Post-surgical reconstruction and prosthetic fitting are crucial for rehabilitation in all procedures, where applicable.

CHAPTER 07

Chapter 7: Errors of Refraction (EOR)

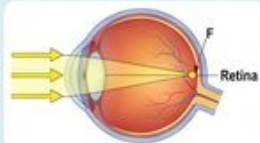


PHYSIOLOGY OF OCULAR REFRACTION & GENERAL INVESTIGATIONS • MYOPIA (SHORT-SIGHTEDNESS) • HYPERMETROPIA (FAR-SIGHTEDNESS) • ASTIGMATISM (NON-SPHERICAL REFRACTIVE ERROR) • APHAKIA, ANISOMETROPIA, AND PRESBYOPIA

PHYSIOLOGY OF OCULAR REFRACTION & GENERAL INVESTIGATIONS

EMMETROPIA

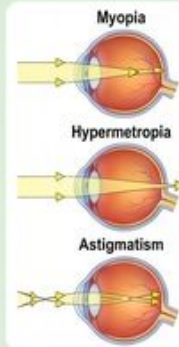
It is the **normal condition** of refraction of the eye in which, with **accommodation completely relaxed**, parallel rays of light coming from infinity **focus precisely on the retina**.



AMETROPIA

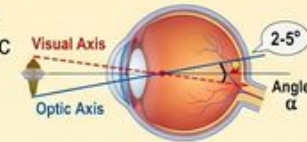
It is the **abnormal condition** of refraction in which, with accommodation **completely relaxed**, parallel rays **do not come to a focus on the retina**.

- The primary types include:
- **Myopia**,
- **Hypermetropia**,
- **Astigmatism**,
- **Anisometropia**, and
- **Aphakia**.



AXES & ANGLES

- **Visual Axis:** The line connecting the **macula** and the **object of regard** passing through the **nodal point**.
- **Optic Axis:** The line connecting the **center** of the **cornea, lens, and retina**.
- **Angle Alpha:** The angle between the visual and optic axes at the nodal point (normally **positive, 2 to 5 degrees**).



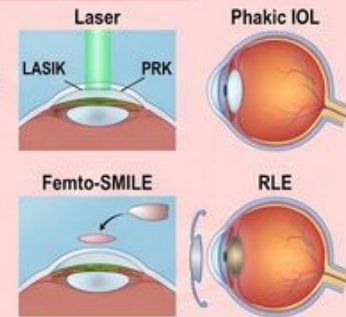
METHODS OF ESTIMATING REFRACTION

- 1) **Retinoscopy:** The **most reliable objective method** using different dioptric powers to neutralize principal meridians. It shows a **scissor reflex** in irregular astigmatism and a **spinning reflex** in keratoconus.
- 2) **Keratometry:** Measures the power and axis of the highest and least meridians of the cornea.
- 3) **Corneal Topography:** The **most sensitive and accurate method** for mapping corneal architecture.



REFRACTIVE SURGERY PRINCIPLES

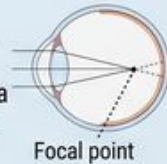
- **Corneal Surgeries:** Adjusting **corneal curvature** using **Excimer laser** (LASIK, PRK) or **Femtosecond laser** (Femto-SMILE for small incision lenticule extraction).
- **Lens Surgeries:** **Phakic IOL** (implanting an ICL over the crystalline lens) or **Refractive Lens Exchange (RLE)**.



MYOPIA (SHORT-SIGHTEDNESS) - An Explanatory Graphic

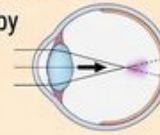
DEFINITION:

A condition where, with accommodation relaxed, parallel rays come to a focus **IN FRONT OF THE RETINA**. The Far Point (Punctum Remotum) is at a finite distance **NEARER THAN INFINITY**, and rays leaving the retina leave the eye convergent.



ROLE OF ACCOMMODATION:

WORSENS MYOPIA by increasing the refractive power of the lens.



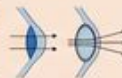
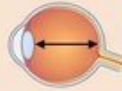
FUNDUS SIGNS OF HIGH MYOPIA:

TESSELLATED (TIGEROID) FUNDUS, chorioretinal degenerations, **POSTERIOR STAPHYLOMA**, **Myopic crescent** (scleral exposure around optic disc), and **FUCHS' SPOT** (black macular area due to subretinal choroidal neovascular membrane hemorrhage).



ETIOLOGY (CAUSES):

- 1) **Axial Myopia** (Most common): Abnormal increase in **AXIAL LENGTH**. Each 1 mm increase results in a **3 DIOPTERS** change.
- 2) **Refractive Myopia**: Increased **REFRACTIVE POWER** via **Curvature myopia** (keratoconus, lenticonus) or **Index myopia** (senile nuclear sclerosis/cataract, or uncontrolled diabetes/hyperglycemia).
- 3) **Abnormal position of the lens**: Anterior dislocation.



TYPES OF AXIAL MYOPIA:

- 1) **Simple**: Starts at puberty (14 yrs), **PROGRESSES SLOWLY**, stabilizes in twenties, rarely exceeds 6D.
- 2) **Progressive (Malignant)**: Starts early (5-10 yrs), **PROGRESSES RAPIDLY OVER** with **DEGENERATIVE FUNDUS CHANGES**.
- 3) **Congenital**: Present at birth, **RAPID PROGRESSION** to 10D then **STATIONARY**, rare complications.



CLINICAL PICTURE & SIGNS:

Symptoms include indistinct **FAR VISION**, **ASTHENOPIA**, photopsia, **MUSCA ARSHITOPHA**, **MUSCA VOLITANTES** (FLOATERS), and defective night vision.

Signs in high myopia include a large globe, deep anterior chamber, sluggish pupil, and **PSEUDO-PROPTOSIS**.



COMPLICATIONS:

Apparent convergent or **TRUE DIVERGENT SQUINT**, Posterior Vitreous Detachment, **RHEGMATOGENOUS RETINAL DETACHMENT**, complicated cataract, pigmentary glaucoma, and **CONSECUTIVE OPTIC ATROPHY**.



TREATMENT:

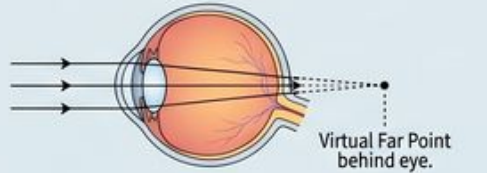
Prescribing the **LEAST TOLERABLE CONCAVE (MINUS) LENSES**, **CONTACT LENSES**, or **REFRACTIVE SURGERIES**.



UNDERSTANDING HYPERMETROPIA (FAR-SIGHTEDNESS)

DEFINITION

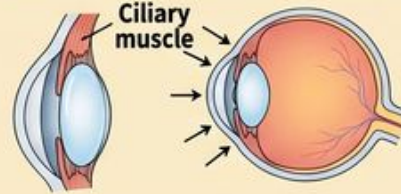
A condition where, with accommodation relaxed, parallel rays come to a focus **BEHIND THE RETINA**.



The **Far Point** is virtual and imaginary behind the eye, and rays leaving the retina leave the eye **divergent**.



ROLE OF ACCOMMODATION



It is the **only refractive error** that can be **actively corrected** by the patient's own ciliary muscle, as **accommodation increases refractive power**.

CLINICAL PICTURE



CHILDREN:
Often **asymptomatic** due to strong accommodation.



ADULTS:
Suffer **progressive visual drop, accommodative asthenopia, and early presbyopia** due to exhaustion.

SIGNS IN HIGH ERRORS:



SMALL GLOBE



SMALL CORNEA



SHALLOW ANTERIOR CHAMBER



NARROW ANGLE



FUNDUS with bright vascular reflex (water-silk), **tortuous vessels**, and **PSEUDO-PAPILDEMA** (blurred disc margins due to crowded nerve fibers).

ETIOLOGY



1) AXIAL:
Failure of normal postnatal eye growth (infants are normally born hypermetropic).



2) REFRACTIVE:
Decreased refractive power via **Curvature hypermetropia** (cornea plana) or **Index hypermetropia** (cortical cataract, or hypoglycemia causing aqueous influx and decreased lens index).



3) POSITIONAL:
Posterior dislocation of the lens.



4) APHAKIA:
Absence of the lens.

COMPONENTS OF HYPERMETROPIA

1) TOTAL:

The full amount measured under **complete cycloplegia** (atropine).

2) LATENT:

The portion constantly corrected by **physiological ciliary tone** (about 1 Diopter).

3) MANIFEST:

The **uncorrected portion** (Total minus Latent)

4) FACULTATIVE:

The part of manifest hypermetropia **correctable by voluntary accommodation**.

5) ABSOLUTE:

The part of manifest hypermetropia that **cannot be corrected** by accommodation.

COMPLICATIONS & TREATMENT



COMPLICATIONS:

Risks include **accommodative esotropia** (true convergent squint) and **primary angle-closure glaucoma**.



TREATMENT:

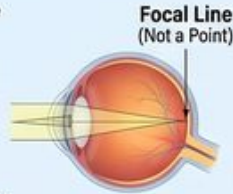
Treated with **SPHERICAL CONVEX (PLUS) LENSES** (full correction for children with esotropia, highest tolerated plus lens for adults), **CONTACT LENSES**, or **REFRACTIVE SURGERY**.

ASTIGMATISM (Non-Spherical Refractive Error)

Understanding the Condition, Causes, and Treatment

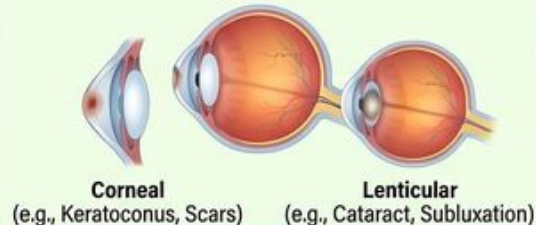
DEFINITION

*A condition where the eye **does not possess the same refractive power** in all meridians. Parallel rays do not focus on a single point but form a focal line. **It cannot be corrected by simple spherical lenses.**



ETIOLOGY

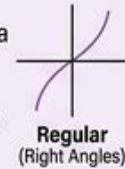
- 1) Corneal (Commonest):** Congenital, or acquired via **keratoconus**, facets, scars, pressure from eyelid tumors/chalazia, or postoperative flattening of the vertical meridian by surgical scars.
- 2) Lenticular:** Early cataract, lens subluxation, or lenticonus.



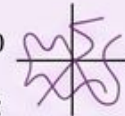
TYPES OF ASTIGMATISM

1) Regular: Principal meridians of highest and lowest power are at right angles, with a gradual power change.

- **Rule of Regular:** Normally, the vertical meridian is more curved than the horizontal (physiological astigmatism $\sim 0.25D$).
- **Sub-classifications:** With-the-rule (vertical is more curved), Against-the-rule (horizontal is more curved), Straight (90/180 degrees), and Oblique.



Regular
(Right Angles)



Irregular
(No Right Angles, Irregular)

2) Irregular: Principal meridians are **not at right angles**, power changes irregularly (e.g., in keratoconus or severe scars), **cannot be corrected by ordinary glasses.**

OPTICAL SUBDIVISIONS OF REGULAR ASTIGMATISM

Simple: One meridian emmetropic, the other ametropic (myopic or hypermetropic).

Compound: Both meridians ametropic of the same type.

Mixed: One meridian myopic, the other hypermetropic.



Simple (One Normal, One Ametropic)



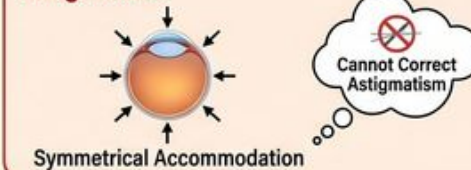
Compound
(Both Same Ametropic)



Mixed (One Myopic, One Hypermetropic)

ROLE OF ACCOMMODATION

Exerted **symmetrically** and circumferentially. Attempting to correct one meridian worsens the other, meaning accommodation **cannot correct astigmatism.**



Symmetrical Accommodation

CLINICAL PICTURE & TREATMENT

Symptoms: Blurred vision, running/dropped letters, tailed points, and severe accommodative asthenopia in low degrees.

Signs: Oval optic disc and distorted Placido disc circles.

Treated using: Cylindrical or spherocylindrical glasses, soft/toric contact lenses, or rigid gas-permeable (RGP) lenses for irregular cases. **Surgical options** include LASIK, PRK, PTK, or keratoplasty.



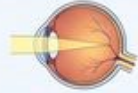
Blurred Vision & Asthenopia



Treatment: Glasses, Contacts, Surgery

APHAKIA

- **Total absence** of the crystalline lens.



Etiology:

- **Congenital** (rare)
- **Acquired** (post-traumatic lens absorption or post-operative extraction)

Optical Problems:

- Severe **hypermetropia**
- **complete loss of accommodation**
- **against-the-rule astigmatism** (from surgical scar contraction)
- **anisometropia**

Signs:

- Extracapsular scar
- **deep anterior chamber**
- **tremulous iris** (iridodonesis)
- **jet-black pupillary area**
- **absence of two of the three Purkinje-Sanson images**

ANISOMETROPIA

- A considerable difference in refraction between the two eyes (*more than 4 Diopters*).



Types:

- **Simple, Compound, or Mixed** anisometropia



Clinical Picture:

Asthenopia and loss of binocular vision resulting in either *alternating vision* or *uniocular vision* leading to **strabismic/anisometropic amblyopia**



Treatment Rule:

Ordinary glasses are strictly contraindicated in unilateral aphakia/anisometropia

because a large plus lens magnifies the retinal image by 30%, causing severe **aniseikonia** and **binocular diplopia**.

Treated with **contact lenses, secondary IOL, or aniseikonic lenses**



PRESBYOPIA

- The physiological age-related **recession of the near point** (*Punctum Proximum*) making near vision uncomfortable. Due to loss of lens elasticity.



Occurs around **45 years** in emmetropes, earlier in hypermetropes, and later in myopes.

Symptoms:

- **Difficulty reading**
- **holding materials further away**
- **accommodative asthenopia**

Management Calculation:

- Correct distant vision first. $\rightarrow 3 \rightarrow D$
- $\rightarrow$ Measure the near point, calculate the total amplitude of accommodation
- $\rightarrow$ keep $\frac{1}{3}$ of it in reserve for comfort
- $\rightarrow$ add **plus lenses** to the remaining $\frac{2}{3}$ so the total refractive power reaches **3 Diopters** (standard reading distance of 33 cm)

Total Power = 3D (at 33cm)

CHAPTER 08

Chapter 8: Glaucoma



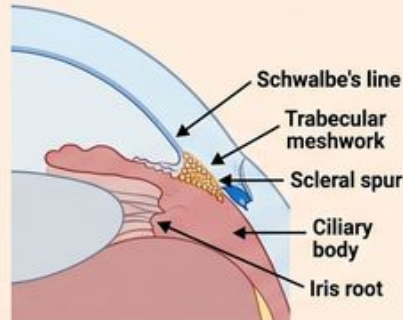
ANATOMY OF THE ANGLE, AQUEOUS PHYSIOLOGY & INTRAOCULAR PRESSURE • GLAUCOMA DEFINITION & CONGENITAL GLAUCOMA (BUPHTHALMOS) • PRIMARY OPEN ANGLE GLAUCOMA (CHRONIC SIMPLE GLAUCOMA) • PRIMARY ANGLE CLOSURE GLAUCOMA (PACG) • SECONDARY GLAUCOMAS (SYSTEMIC, LOCAL, AND PHACOGENIC)

Anatomy of the Anterior Chamber Angle

It is the angle between the **peripheral cornea** and the **iris root**, examined clinically using **Gonioscopy**.

Gonioscopic structures (from anterior to posterior) include **Schwalbe's line** (end of Descemet's membrane), **Trabecular meshwork** (sponge-like bands), **Scleral spur** (origin of ciliary muscle), anterior surface of **Ciliary body**, and the root of the iris.

Schlemm's canal and aqueous veins are hidden deep within the limbus.



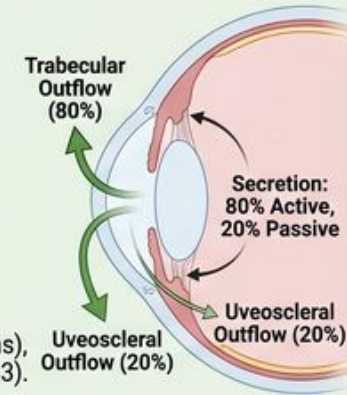
Physiology of the Aqueous Humor

A transparent fluid (**1.25 ml**) secreted by the **ciliary body** processes.

Formation is **80% active secretion** (via carbonic anhydrase) and **20% passive ultrafiltration**.

It flows from the posterior chamber through the pupil into the anterior chamber, exiting via **Trabecular outflow** (80%) and **Uveoscleral outflow** (20%).

Functions include regulating IOP, providing nutrition to avascular structures (cornea/lens), and serving as a refractive medium (RI = 1.33).



Intraocular Pressure (IOP)

Normal IOP ranges from **10 to 21 mmHg** (average 16 mmHg).

It exhibits a **diurnal variation** (highest in the morning, lowest in the evening by 2-3 mmHg).

It is measured via **Tonometry**:

Indentation (Schiotz tonometer, affected by scleral rigidity),

Applanation (Goldmann tonometer, the gold standard, flattens 3.06 mm of cornea), and **Non-contact Air-puff** tonometer.

Tonometry Methods



Indentation
(Schiotz)



Applanation
(Goldmann - Gold Standard)



Non-contact
Air-puff

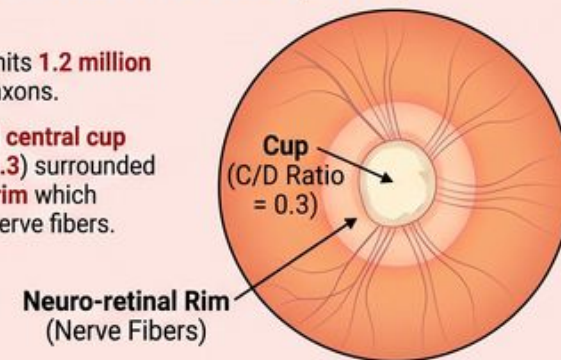


10-21 mmHg
(Normal Range)

Optic Nerve Head (ONH)

The optic disc transmits **1.2 million** retinal ganglion cell axons.

It has a physiological **central cup** (normal **C/D ratio = 0.3**) surrounded by the **neuro-retinal rim** which contains the actual nerve fibers.

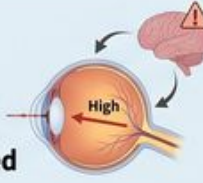


Glaucoma Definition & Congenital Glaucoma (Buphthalmos)

A Progressive Optic Neuropathy

• Definition & Classification

Glaucoma is a **progressive optic neuropathy** characterized by **optic nerve damage** and **visual field defects**, with **elevated IOP** being the major risk factor. Classified into **Congenital/Developmental** and **Acquired** (Primary Open/Closed angle, and Secondary).



• Congenital Glaucoma (Primary Infantile / Buphthalmos)

Caused by **trabeculo-dysgenesis** (failure of mesodermal atrophy, abnormal anterior iris insertion, or absent Schlemm's canal). Symptoms present between **birth and the first year** as a classic triad: **Lacrimation, Photophobia, and Blepharospasm** due to corneal nerve stretching.



• Clinical Signs

1) Cornea

Edema, progressive enlargement (>12 mm diameter), and **Haab's striae** (horizontal tears in Descemet's membrane).



3) Anterior Chamber

Abnormally deep.



5) Optic Lens

Enlarged eyeball with a **bluish sclera** due to thinning.



4) Refraction

Axial myopia



• Clinical Signs

1) Cornea

Edema, progressive enlargement (>12 mm diameter), and **Haab's striae** (horizontal tears in Descemet's membrane).



2) Globe & Sclera

Enlarged eyeball with a **bluish sclera** due to thinning.



3) Anterior Chamber

Abnormally deep.



4) Iris & Lens

Tremulous, atrophic iris with possible **lens subluxation**.



5) Optic Disc

Glaucomatous **cupping** is a **late sign** due to the **elastic expansibility** of the infant's globe.



6) Refraction

Axial myopia.



• Differential Diagnosis



Megalocornea, congenital cloudy cornea (rubella, birth trauma, mucopolysaccharidosis), and **secondary infantile glaucoma** (retinoblastoma, PHPV, ROP).

• Management

Strictly surgical. Medical treatment is only used **temporarily** to clear corneal edema. Surgeries include **Goniotomy** (if cornea is clear), **Trabeculotomy** (if cornea is hazy), and in resistant cases, **Subscleral Trabeculectomy, Valve implantation** (Ahmed valve), or **Cyclodestructive procedures**.

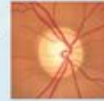
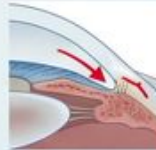


Primary Open Angle Glaucoma (Chronic Simple Glaucoma)

Definition & Etiology

A bilateral, asymmetrical, **non-congestive elevation of IOP** in the presence of an **open angle**, leading to **optic nerve damage** and visual field loss.

Caused by **age-related sclerosis** of the **trabecular meshwork**. Risk factors include age >50, positive family history, **dark races**, myopia, diabetes, and vasospastic disorders.

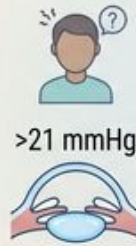


Clinical Picture

Usually **asymptomatic** until late stages. May present with **mild headache**, early presbyopia, or **night blindness**.

Signs include **elevated IOP** (>21 mmHg) with an **open angle** on gonioscopy.

Normal-tension glaucoma exists, requiring evaluation of diurnal variation (>4 mmHg) and tonography.



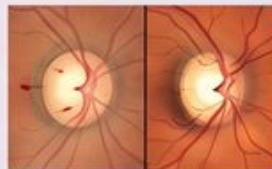
>21 mmHg



Glaucomatous Optic Cupping

Early signs include **enlarged C/D ratio** (>0.4), **vertical elongation** of the cup, cup **asymmetry** between eyes, and **splinter hemorrhages**.

Late signs show a **deep cup** (>0.7), overhanging undermined edges, interruption of retinal vessels, and **nasal shift** of blood vessels.



Early

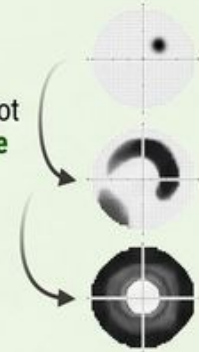
Late

Visual Field Changes

Central defects include **Paracentral scotoma** in Bjerrum area, **Seidie scotoma** (sickle-shaped blind spot extension), **Arcuate (Bjerrum) scotoma**, **Baring of the blind spot**, and **Ronne nasal step**.

Peripheral changes lead to **concentric contraction** and eventually a **tubular field**.

Macular vision is preserved until the very end.



Management

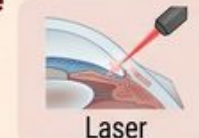
1) **Medical (Life-long)**: First line includes topical **Prostaglandin analogues** (increase uveoscleral outflow) and **Beta-blockers** (decrease aqueous formation). Second line includes **Carbonic Anhydrase Inhibitors (CAI)** and Alpha-2 agonists.

2) **Laser Trabeculoplasty (Argon/YAG)**: Applied to the trabecular meshwork to **increase outflow**.

3) Surgical: **Trabeculectomy** (with antimetabolites like Mitomycin C) if medical therapy fails, **Seton valves** for refractory cases, or **Cyclodestructive procedures**.



Meds



Laser



Surgery/Seton

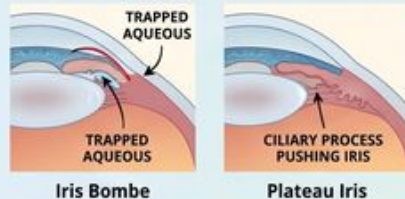
PRIMARY ANGLE CLOSURE GLAUCOMA (PACG)

PREDISPOSING FACTORS & PATHOGENESIS

Typically occurs **bilaterally** in **neurotic females** over **40 years** with **hypermetropic** (small) eyes, **shallow anterior chambers**, and **thick lenses**.



Pathogenesis involves either **Pupillary Block** (iris-lens apposition traps aqueous, causing **iris bombe** that occludes the angle) or **Plateau Iris** (anteriorly positioned **ciliary processes** push the peripheral iris forward).



CLINICAL STAGES

1) **Latent Stage:** Asymptomatic, **shallow** anterior chamber. Treated prophylactically with **laser peripheral iridotomy**.



2) **Intermittent (Prodromal) Stage:** Rapid, spontaneously **reversible** angle closure triggered by **mydriasis** (darkness, stress). Presents with **transient blurred vision**, **frontal headache**, and **rainbow haloes** around lights. Treated by iridotomy to prevent acute attacks.



3) **Acute Congestive Phase:** Sudden, total angle closure. **Severe periocular pain**, **sudden drop of vision** (down to hand movement), **nausea**, and **vomiting**. Signs include **intensely high IOP** (up to 50 mmHg), **stony hard globe**, **severe corneal edema**, circumcorneal ciliary injection (CCCI), and a characteristic **vertically oval, fixed, semi-dilated bluish pupil**.



4) **Chronic Phase:** Follows untreated acute attacks. **Permanent Peripheral Anterior Synechiae (PAS)** form, causing **progressive cupping** and **field loss** resembling POAG.



5) **Absolute Phase:** End-stage blind painful eye with **total optic atrophy** and eventual **ciliary shutdown**.



MANAGEMENT OF ACUTE ATTACK

Requires **urgent hospitalization**. **Medical therapy lowers IOP first:** **IV Hyperosmotics** (Mannitol 20%), **Systemic CAI** (Acetazolamide), **topical Beta-blockers**, and **intensive Miotics** (Pilocarpine 2%).



Once IOP drops and the cornea clears, **surgical/laser Peripheral Iridectomy is mandatory** for the affected eye, and **prophylactic iridotomy** must be performed on the **fellow eye**.



Understanding Secondary Glaucomas (Systemic, Local, and Phacogenic)

SYSTEMIC CAUSES (Raised Episcleral Venous Pressure)



Conditions like **Superior Vena Cava Obstruction** and **Dysthyroid Ophthalmopathy** lead to increased episcleral venous pressure, hindering aqueous outflow.

INTRAOCULAR INFLAMMATION (UVEITIS)



Acute Iridocyclitis: Causes **OPEN-ANGLE GLAUCOMA** due to inflammatory cells, plasmoid aqueous, and trabecular edema.



Chronic Iridocyclitis: Causes **CLOSED-ANGLE GLAUCOMA** through extensive Peripheral Anterior Synechiae (PAS) or 360-degree ring posterior synechia (**seclusio pupillae**) leading to iris bombe.

LENS-INDUCED (PHACOGENIC) GLAUCOMAS



1) Phacolytic Glaucoma: Hypermature cataracts leak lens proteins, engulfed by **macrophages** which obstruct the trabecular meshwork.



2) Phacoanaphylactic Glaucoma: Autoimmune **granulomatous reaction** against exposed lens proteins following capsular rupture (trauma/surgery).



3) Phacomorphic Glaucoma: Intumescent senile cortical cataract; swollen, water-logged lens precipitates **acute secondary angle closure**.



4) Lens Displacement: Anterior dislocation causes severe **pupillary block**.



5) Exfoliation Glaucoma: True exfoliation (infrared injury) or **Pseudo-exfoliation** (dandruff-like fibrillar deposits on lens, ciliary processes, and angle).

NEOVASCULAR GLAUCOMA



Severe, intractable glaucoma from fibrovascular membrane growth (**rubeosis iridis**) over iris and angle, secondary to retinal ischemia (**CRVO "100-day glaucoma"**, Proliferative Diabetic Retinopathy, or intraocular tumors).

TRAUMA & STEROID-INDUCED



Blunt Trauma: Can tear the ciliary body (**Angle recession glaucoma**) or cause **hyphema**.



Prolonged Topical Corticosteroid Therapy: Induces trabecular mucopolysaccharide deposition in susceptible individuals, creating a clinical picture identical to **primary open-angle glaucoma**.

CHAPTER 09

Chapter 9: Uveitis



ANATOMY AND PHYSIOLOGY OF THE UVEAL TRACT • CLASSIFICATION & ETIOLOGY OF UVEITIS (IRIDOCYCLITIS) • CLINICAL PICTURE & SIGNS OF ACUTE ANTERIOR UVEITIS (IRIDOCYCLITIS) • COMPLICATIONS AND SEQUELAE OF IRIDOCYCLITIS • MANAGEMENT OF IRIDOCYCLITIS & SYMPATHETIC OPHTHALMIA

ANATOMY AND PHYSIOLOGY OF THE UVEAL TRACT

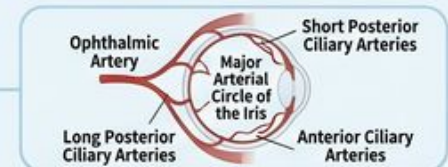
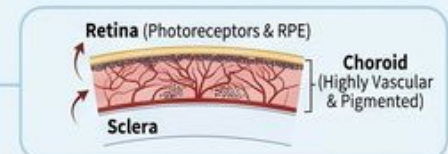
DEFINITION & COMPONENTS: The **uveal tract** is the **middle vascular and pigmented coat** of the eyeball. It consists of three contiguous parts from anterior to posterior: the **Iris**, the **Ciliary Body**, and the **Choroid**.

THE IRIS: The **anterior-most part**, forming a movable vertical diaphragm between the anterior and posterior chambers, with a central circular opening (the **pupil**). It **controls the amount of light** entering the eye via two muscles: the **sphincter pupillae** (parasympathetic supply, causes **miosis/constriction**) and the **dilator pupillae** (sympathetic supply, causes **mydriasis/dilation**).

THE CILIARY BODY: Extends from the iris root to the **ora serrata**. It is divided into the **pars plicata** (the anterior highly vascular part containing **ciliary processes** that actively **secrete aqueous humor**) and the **pars plana** (the posterior flat part). It contains the **ciliary muscle**, which is entirely responsible for the mechanism of **accommodation**.

THE CHOROID: The **highly vascular posterior portion** of the uveal tract, extending from the ora serrata to the optic nerve. It **nourishes the outer layers of the retina** (photoreceptors and RPE) and acts as a **dark chamber to absorb excess scattered light** via its heavy pigmentation.

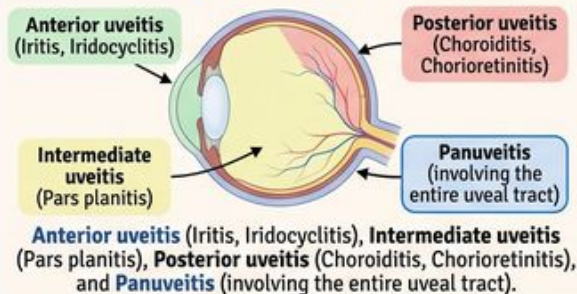
BLOOD SUPPLY: The uveal tract is richly supplied by the **ophthalmic artery** via the **short posterit** (supplying the choroid), the **long posterior ciliary arteries**, and the **anterior ciliary arteries** (which anastomose to form the **major arterial circle of the iris** to supply the iris and ciliary body).



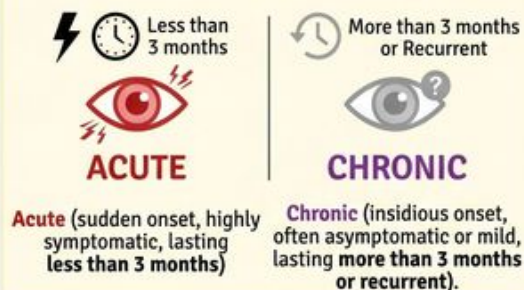
CLASSIFICATION & ETIOLOGY OF UVEITIS (Iridocyclitis)

CLASSIFICATION OF UVEITIS

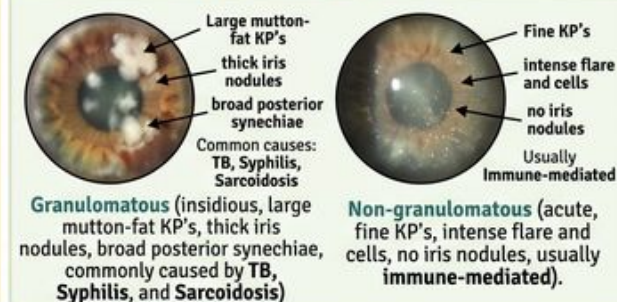
1) Anatomical:



2) Clinical Course:

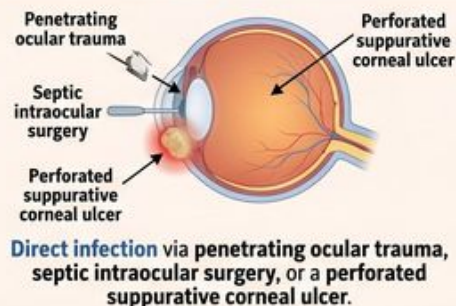


3) Pathological:

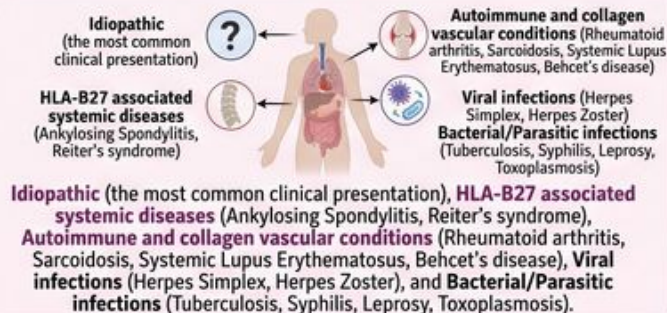


ETIOLOGY OF ANTERIOR UVEITIS (Iridocyclitis)

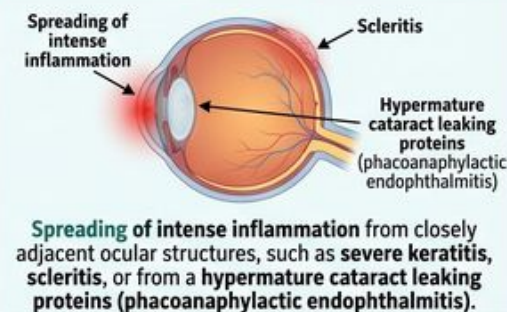
- Exogenous:



- Endogenous (Systemic & Immune):



- Secondary:



CLINICAL PICTURE & SIGNS OF ACUTE ANTERIOR UVEITIS (IRIDOCYCLITIS)

CLINICAL SYMPTOMS

Acute, deep, throbbing ocular pain (typically worse at night and exacerbated upon accommodation)



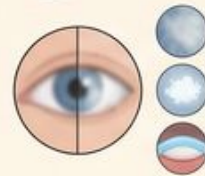
Severe photophobia



Reflex lacrimation



Marked blurring of vision (due to aqueous turbidity, exudates, and corneal edema)



CLINICAL SIGNS (Anterior Segment Examination)

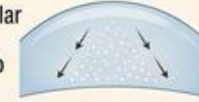
1) Ciliary Injection

A deep, violaceous, circumcorneal vascular flush around the limbus.



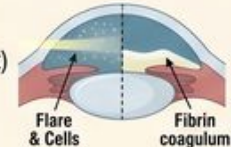
2) Cornea

Endothelial dusting and Keratic Precipitates (KP's). KP's are cellular deposits of leukocytes on the corneal endothelium, classically arranged in a base-down triangular pattern (Arlt's triangle) due to anterior chamber convection currents.



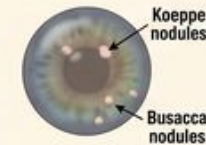
3) Anterior Chamber

Shows **Aqueous Flare** (due to protein leakage from a broken blood-aqueous barrier, producing a visible optical Tyndall effect) and **Aqueous Cells** (active circulating lymphocytes). Severe exudation may settle dependently at the bottom to form a sterile **Hypopyon**, or rich fibrin may form a solid coagulum.



4) Iris Changes

The iris becomes edematous, water-logged, muddy, and loses its normal crisp crypt pattern. Inflammatory cellular aggregates may appear at the pupillary margin (**Koepe nodules**) or within the stroma (**Busacca nodules**).



5) Pupil

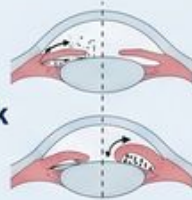
Initially constricted (**Miosis**) due to irritative sphincter spasm. Its reaction to light becomes extremely sluggish, and it may acquire an irregular festooned shape due to the formation of **Posterior Synechiae** (inflammatory adhesions between the posterior iris surface and the anterior lens capsule).



Complications and Sequelae of Irdocyclitis

SECONDARY GLAUCOMA

Can occur via **dual mechanisms**. In the **early acute phase** (Open-angle), it is due to **mechanical blockage** of the **trabecular meshwork** by inflammatory cells, thick fibrin, or high-protein plasmoid aqueous.



In the **late chronic phase** (Closed-angle), it is due to organized **Peripheral Anterior Synechiae (PAS)**, or a complete 360-degree **Ring Posterior Synechia (Seclusio Pupillae)**.

Seclusio pupillae physically blocks aqueous flow, causing the peripheral iris to bow forward (**Iris Bombe**), leading to bow forward (**Iris Bombe**), leading to severe **secondary angle-closure glaucoma**.

CYCLITIC MEMBRANE

The **organization of rich inflammatory exudates** behind the lens into a **dense contractile fibrous membrane**, which can pull on the ciliary body causing **ciliary detachment** and **profound hypotony**.



Ciliary Detachment & Hypotony

COMPLICATED CATARACT

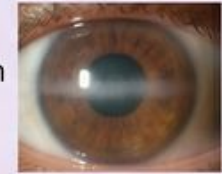
A classic **posterior subcapsular cataract** developing due to **altered aqueous metabolic nutrition**, **diffusing inflammatory toxins**, and the **prolonged therapeutic use of topical corticosteroids**.



Posterior Subcapsular Cataract

BAND KERATOPATHY

A **progressive horizontal band of grayish-white calcium salt deposition** in **Bowman's membrane** of the cornea, typically occurring in **chronic smoldering uveitis**, especially in children with **Juvenile Idiopathic Arthritis (JIA)**.



Calcium Salt Deposition in Bowman's Membrane

MACULAR COMPLICATIONS

Cystoid Macular Edema (CME) and **macular ischemia**, which stand as the **major leading causes of permanent visual loss** in chronic uveitis patients.



Cystoid Macular Edema & Ischemia

PHTHISIS BULBI

The **devastating end-stage of severe chronic destructive uveitis**; it presents as a **shrunken, totally disorganized, soft, and blind eye** due to **permanent failure of aqueous secretion** by the destroyed, fibrosed ciliary body.



Shrunken, Disorganized, Blind Eye

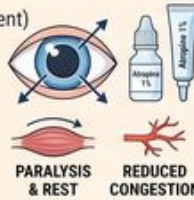
MANAGEMENT OF ACUTE IRIDOCYCLITIS

COMPREHENSIVE THERAPEUTIC STRATEGIES

1. MYDRIATICS & CYCLOPLEGICS (Atropine 1% drops/ointment)

THE ABSOLUTE MOST CRITICAL LOCAL THERAPY.

It forcibly breaks existing posterior synechiae and prevents new ones, completely paralyzes the ciliary muscle and sphincter pupillae to provide absolute rest and relieve painful spasm, and directly decreases ciliary vascular congestion.

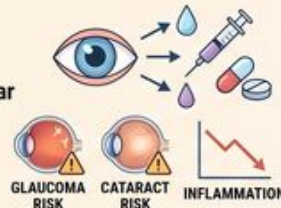


2. CORTICOSTEROIDS

POTENT INFLAMMATION SUPPRESSION & VASCULAR EXUDATION REDUCTION.

Administered **topically** (drops/ointments), via **periocular injections** (subconjunctival or sub-Tenon's for severe cases), or **systemically** in severe bilateral cases.

Patients must be **STRICTLY MONITORED** for steroid-induced **glaucoma** and **cataract**.

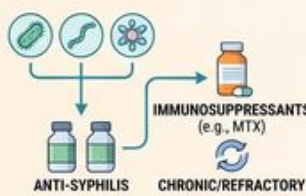


3. SPECIFIC & SYSTEMIC THERAPY

TREATING THE UNDERLYING SYSTEMIC CAUSE.

Examples include **anti-tuberculous drugs** and **anti-syphilitic antibiotics**.

SYSTEMIC IMMUNOSUPPRESSANTS (e.g., Methotrexate, Cyclosporine) are reserved for **refractory** or **autoimmune chronic** cases.

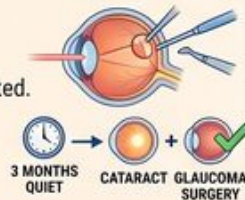


4. SURGICAL INTERVENTION

RELIEVING PUPILLARY BLOCK & IRIS BOMBE.

Laser Peripheral Iridotomy or surgical Iridectomy is indicated.

CATARACT OR GLAUCOMA SURGERIES are performed **only** after the eye is **QUIET AND INFLAMMATION-FREE FOR 3 MONTHS**.



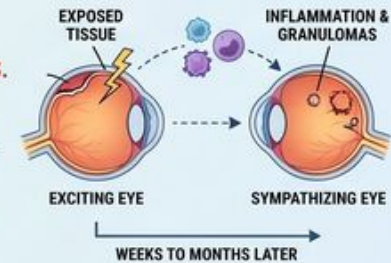
SYMPATHETIC OPHTHALMIA

RARE, DEVASTATING BILATERAL INFLAMMATION

DEFINITION

A RARE, DEVASTATING, BILATERAL, CHRONIC GRANULOMATOUS PANUVEITIS.

It exclusively follows a **penetrating ocular trauma** or **intraocular surgery** (typically involving iris/uveal prolapse) in one eye (the **exciting eye**), leading to a **delayed autoimmune inflammatory response** against exposed uveal and retinal antigens in the normal, uninjured fellow eye (the **sympathizing eye**) **weeks to months later**.

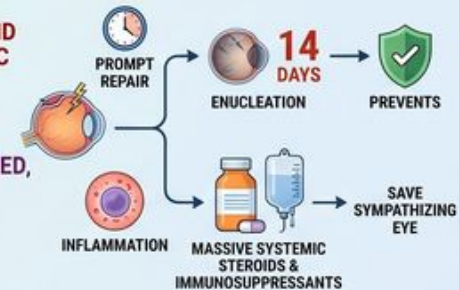


PREVENTION & TREATMENT

PROMPT, METICULOUS MICROSURGICAL REPAIR of penetrating injuries.

If the injured eye is **COMPLETELY BLIND AND DISORGANIZED**, **PROPHYLACTIC ENUCLEATION WITHIN 14 DAYS OF TRAUMA ABSOLUTELY PREVENTS THE CONDITION**.

Once the disease is **FULLY ESTABLISHED**, it requires **MASSIVE, PROLONGED SYSTEMIC CORTICOSTEROIDS AND IMMUNOSUPPRESSIVE THERAPY** to save the sympathizing eye.



IMPORTANT: EARLY RECOGNITION AND INTERVENTION ARE CRUCIAL IN BOTH CONDITIONS TO PREVENT IRREVERSIBLE VISION LOSS.

CHAPTER 10

Chapter 10: Strabismus

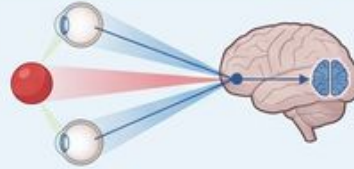


PHYSIOLOGY OF BINOCULAR VISION & APPARENT SQUINT • LATENT SQUINT (HETEROPHORIA) • MANIFEST PARALYTIC SQUINT (INCOMITANT STRABISMUS) • MANIFEST CONCOMITANT SQUINT (HETEROTROPIA) • EXAMINATION & TREATMENT OF CONCOMITANT SQUINT

PHYSIOLOGY OF BINOCULAR VISION & APPARENT SQUINT

BINOCULAR VISION

The **coordinated use of both eyes** to produce a single **visual mental impression**.



Develops by **6 MONTHS** (maintained fixation, ) and is **FULLY ESTABLISHED** with **FUSION** by **6 YEARS** ()

- Essential for **STEREOPSIS** (Depth Perception, )
- ENLARGING** the **VISUAL FIELD** ()
- and **MASKING** OPTICAL DEFECTS ()

GRADES OF BINOCULAR VISION

1) GRADE I: SIMULTANEOUS PERCEPTION

Ability to see **TWO DISSIMILAR OBJECTS SIMULTANEOUSLY**.



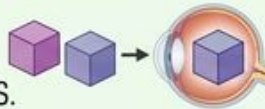
2) GRADE II: FUSION

Ability to **SUPERIMPOSE TWO INCOMPLETE SIMILAR OBJECTS** to form a **COMPLETE PICTURE**.



3) GRADE III: STEREOSCOPIC VISION

Ability to **APPRECIATE 3D DEPTH** by superimposing **SLIGHTLY DISSIMILAR OBJECTS**.



BASIC DEFINITIONS

ORTHOPHORIA: Perfect muscle balance with **PARALLEL VISUAL AXES** even when binocular vision is **ABOLISHED**.



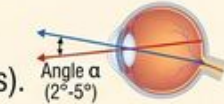
VISUAL AXIS: Line connecting the **MACULA** to the **OBJECT** through the **NODAL POINT**.



OPTIC AXIS: Line connecting the **CENTERS** of the **CORNEA, LENS, and RETINA**.



ANGLE ALPHA: The angle between the **VISUAL** and **OPTIC AXES** (normally **POSITIVE**, 2 to 5 degrees).



APPARENT SQUINT (PSEUDO-STRABISMUS)

Appearance of ocular deviation but **VISUAL AXES ARE COMPLETELY NORMAL**.

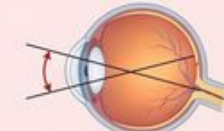
APPARENT CONVERGENT SQUINT:

Caused by **EPICANTHUS SKIN FOLDS** covering the **INNER CANTHUS**.



APPARENT DIVERGENT SQUINT:

Caused by a **LARGE POSITIVE ANGLE ALPHA** in **HIGH HYPERMETROPIA**.



LATENT SQUINT (Heterophoria)

Definition

Ocular deviation with an abnormal direction of the visual axes that is only manifested when **binocular vision is dissociated** (e.g., by fatigue or covering one eye).



Types

- **Esophoria** (tendency to deviate inwards)
- **Exophoria** (outwards)
- **Hyperphoria** (upwards)
- **Hypophoria** (downwards)



Etiology

Uncorrected errors of refraction (**Hypermetropia** causes esophoria due to **excessive accommodation**; **Myopia** causes exophoria) or mild **weakness of extraocular muscles**.



Clinical Picture

Usually compensated and asymptomatic.

If **decompensated** (due to severe fatigue or ill-health), it presents with **muscular asthenopia** (eye strain after prolonged close work, **ired**), blurring, and running of letters.



Diagnostic Tests

(Dissociation of Binocular Vision)

- 1) **Cover Test:** Covering one eye abolishes binocular vision; the covered eye deviates. Upon removing the cover, it rapidly returns to its original straight position.
- 2) **Maddox Rod Test:** Dissociates vision by changing the retinal image appearance to detect distance latent squint.
- 3) **Maddox Wing Test:** Used to perfectly measure near latent squint.



Treatment

Compensated cases require no treatment.

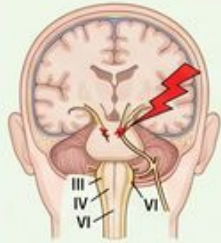


Decompensated cases require: **exact correction of refractive errors**, **Orthoptic exercises** (e.g., pencil-to-nose for convergence insufficiency, **relieving prisms** (base directed towards the deviation), or **surgical muscle correction** in highly resistant cases.

MANIFEST PARALYTIC SQUINT (INCOMITANT STRABISMUS)

Understanding Ocular Deviation from Muscle Dysfunction

DEFINITION & ETIOLOGY



Ocular deviation due to complete (**PARALYSIS**) or partial (**PARESIS**) loss of muscle function.

The angle of squint is **HIGHLY VARIABLE** in different gaze directions.

Caused by lesions affecting **OCULAR MOTOR NUCLEI, CRANIAL NERVES, or MUSCLES.**

CLINICAL SIGNS



OCULAR DEVIATION occurs in the **OPPOSITE** direction of the paralyzed muscle.



SECONDARY DEVIATION is **GREATER** than Primary Deviation (normal eye deviates excessively when paralyzed eye fixes).



MARKED LIMITATION of ocular movements in the direction of the paralyzed muscle.

CLINICAL SYMPTOMS



1) **DIPLOPIA**: Double vision **DISAPPEARS COMPLETELY** when one eye is covered. Can be **UNCROSSED** (abducens palsy) or **CROSSED** (oculomotor palsy).



2) **FALSE PROJECTION** (Past Pointing): Incorrect spatial orientation of objects in the field of the paralyzed muscle due to **EXCESSIVE BRAIN INNERVATION.**



3) **OCULAR TORTICOLLIS**: Compensatory **HEAD POSTURE** (face turn, head tilt, chin elevation) to avoid diplopia.



4) **VERTIGO & NAUSEA**: Secondary to severe spatial diplopia.

MANAGEMENT



TREATING the underlying neurological or systemic cause.



TEMPORARY OCCLUSION of the paralyzed eye to avoid diplopia.



RELIEVING PRISMS to align images.



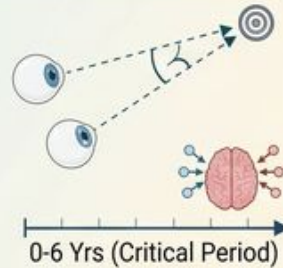
SURGICAL CORRECTION is **ONLY** considered if no spontaneous recovery occurs **AFTER 6 MONTHS.**

Manifest Concomitant Squint (Heterotropia): An Explanatory Graphic

Definition & Etiology

Ocular deviation where the angle of squint remains **constant in all directions of gaze**.

Caused by **sensory, motor, or central fusion obstacles** during the **first 6 years of life** (the critical period of flux).



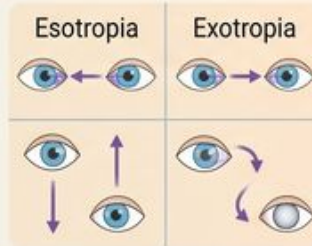
Common Causes

1) Refractive Errors
Hypermetropia leads to **excessive accommodation** and **convergence** (Accommodative Esotropia).

2) Stimulus Deprivation
Severe vision drop in one eye due to congenital ptosis or **dense cataract** disrupts normal fusion.

Common Causes

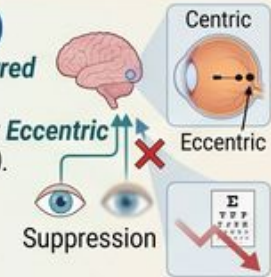
Hypermetropia leads to **excessive accommodation** and **convergence** (Accommodative Esotropia).
Myopia leads to **under-accommodation** and **divergent squint** (Exotropia).



Sensory Adaptations (Suppression)

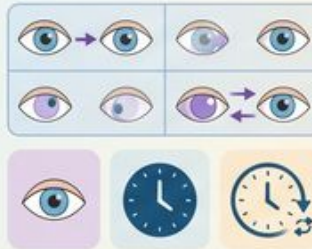
To **avoid diplopia**, the brain actively **inhibits the blurred image** from the squinting eye.

- Types of Fixation: **Centric** (using the normal fovea) or **Eccentric** (an extrafoveal retinal point takes over foveal function).
- **Strabismic Amblyopia** (Amblyopia Ex-anopsia): **Permanent profound drop in visual acuity** in the unilaterally suppressed eye.



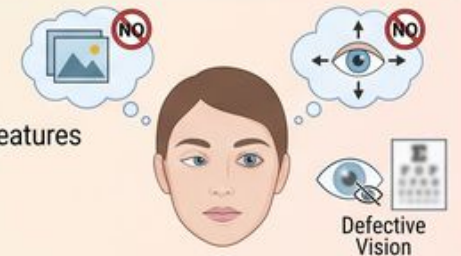
Classifications

Can be **Convergent** (Esotropia), **Divergent** (Exotropia), **Vertical**, or **Mixed**. It can be **Unilateral** (one eye always squints) or **Alternating** (either eye takes up fixation alternately). It may also be **Constant** or **Intermittent**.



Clinical Picture

Noticeable ocular deviation and **defective vision**, but classically features **NO diplopia** and **NO limitation** of extraocular movements.



EXAMINATION & TREATMENT OF CONCOMITANT SQUINT

DIAGNOSTIC EXAMINATION

1) Cover / Uncover Test:

Differentiates LATENT from MANIFEST squint. In concomitant squint, PRIMARY and SECONDARY angles of deviation are PERFECTLY EQUAL (unlike paralytic squint). It also detects alternating vs. unilateral fixation.



2) Hirschberg Test (Corneal Reflex):

The patient fixes on a light at 0.5m. The reflex is DECENTERED in the squinting eye. A reflex at the PUPILLARY MARGIN = 15 degrees angle; MIDWAY to LIMBUS = 30 degrees; AT THE LIMBUS = 45 degrees.



3) Synoptophore (Major Amblyoscope):

The MOST ACCURATE method to measure the EXACT SUBJECTIVE and OBJECTIVE angle of squint.



4) Prism Bar Cover Test:

PRISMS of increasing power are placed until the CORNEAL REFLEX is PERFECTLY CENTRALIZED.



COMPREHENSIVE TREATMENT LINES

- **Timing:** Must be initiated AS EARLY AS POSSIBLE to prevent permanent AMBLYOPIA and allow BINOCULAR VISION development.



1) **Optical:** ACCURATE CORRECTION of any REFRACTIVE ERROR (essential for ACCOMMODATIVE ESOTROPIA).



2) **Occlusion Therapy:** PATCHING the SOUND FIXING EYE to force the brain to use the AMBLYOPIC SQUINTING EYE.



3) **Pleoptics:** SPECIALIZED INTENSE STIMULATION of the TRUE FOVEA in cases of ECCENTRIC FIXATION.



4) **Orthoptics:** EXERCISES to RESTORE FUSION and STEREOPSIS.



5) Surgical Correction:

RECESSING (weakening) overacting muscles or RESECTING (strengthening) underacting muscles to MECHANICALLY ALIGN the visual axes.



CHAPTER 11

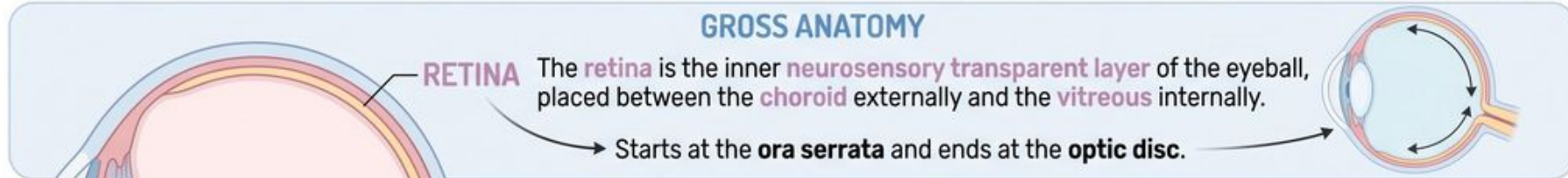
Chapter 11: Retina



ANATOMY OF THE RETINA, BLOOD SUPPLY, AND THE VITREOUS • DIABETIC RETINOPATHY (DR) & HYPERTENSIVE RETINOPATHY • RETINOPATHY OF PREMATURITY (ROP) & RETINAL VEIN OCCLUSIONS • CENTRAL AND BRANCH RETINAL ARTERY OCCLUSION (CRAO & BRAO) • RETINAL DETACHMENT (RD): RHEGMATOGENOUS, TRACTIONAL, AND EXUDATIVE • RETINITIS PIGMENTOSA (PRIMARY PIGMENTARY RETINAL DYSTROPHY)

ANATOMY OF THE RETINA, BLOOD SUPPLY, AND THE VITREOUS

GROSS ANATOMY



MICROSCOPIC ANATOMY (10 Layers from Outside Inwards)

- 1) **Retinal pigment epithelium (RPE)**: Single layer of pigmented cubical cells.
- 2) **Photoreceptors: Rods and cones**, the light-sensitive layer.
- 3) Outer limiting membrane.
- 4) **Outer nuclear layer**: Nuclei of **rods and cones**.
- 5) Outer plexiform layer (synapses).
- 6) **Inner nuclear layer**: **Bipolar and Muller cells**.
- 7) Inner plexiform layer.
- 8) **Ganglion cell layer**.
- 9) **Nerve fiber layer**: Axons of ganglion cells forming the **optic nerve**.
- 10) Inner limiting membrane.

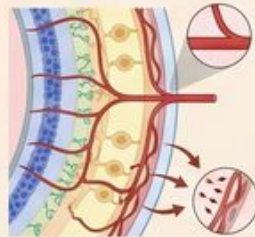
THE MACULA & FOVEA

MACULA LUTEA
(dark central area **3 mm** temporal to the optic disc)

FOVEA
(central depression)

FOVEOLA
(its completely **avascular center**)

BLOOD SUPPLY

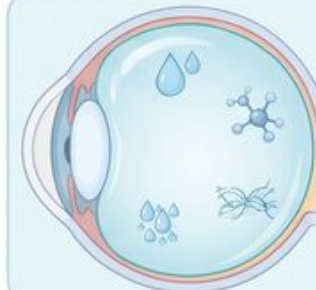


The **inner 5 layers** are supplied by the **central retinal artery** (branch of ophthalmic artery).

The **outer 5 layers** are completely **avascular** and nourished by **diffusion** from the **choriocapillaris** of the **choroid**.

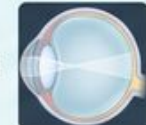
(In 10-20% of the population, a **cilioretinal artery** exists to supply the macula).

THE VITREOUS



A clear **gel** occupying **2/3** of the globe, formed of **99% water**, **hyaluronic acid**, **fine collagen network**, and few cells.

Stabilizes globe volume, acts as a **cushion** for the retina, and serves as an **optical medium**.



Diabetic & Hypertensive Retinopathy: Mechanisms & Signs

DIABETIC RETINOPATHY (DR)

Overview & Risk Factors

Diabetic Retinopathy (DR): The **commonest cause of blindness** in the working population.

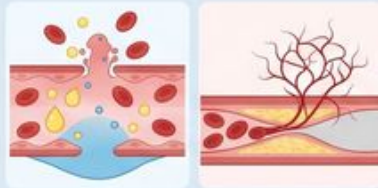
Risk factors include young age, **long duration**, **poor control**, **coexisting hypertension**, and pregnancy.



DR Pathogenesis: Micro-vessel Damage

DR Pathogenesis: Affects micro-vessels leading to:

- 1) **Microvascular leakage**
(decrease in pericytes leads to **breakdown of blood-retinal barrier**, causing **hemorrhages**, **edema**, and **hard lipid exudates**).
- 2) **Microvascular occlusion**
(progressive thickening of basement membrane and platelet aggregation leads to **ischemia**, releasing vaso-formative substances that cause **new vessel formation**).



HYPERTENSIVE RETINOPATHY

General Progression

Hypertensive Retinopathy: **Mild hypertension** leads to **arteriosclerosis** and **generalized arteriolar narrowing**.

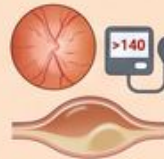
Sustained hypertension damages vessel walls causing **vascular leakage** (**flame-shaped hemorrhages**, **retinal edema**, **hard exudates**).

Severe hypertension leads to precapillary closure and **cotton wool spots** (ischemia).



Special & Severe Forms

Malignant Hypertension: **Diastolic BP > 140 mmHg** damages choroidal arterioles, leading to the hallmark signs of **severe papilledema** and **massive exudative retinal detachment**.



Retinopathy of Toxemia of Pregnancy: Occurs **late in pregnancy** (after 20th week). Fundus shows **severe arteriolar spasm**, **macular star**, flame-shaped hemorrhages, **cotton wool spots**, papilledema, and massive exudation that may produce **serous retinal detachment**. **No arteriosclerosis** is seen because the hypertension is not longstanding. Treatment requires **urgent elective termination of pregnancy** to save the mother's vision and life.





Retinopathy of Prematurity (ROP)

A proliferative retinopathy affecting **pre-term infants** (<30 weeks gestation) and **low birth weight** (<1500 gm) who are exposed to **high oxygen therapy**, which damages the incompletely vascularized retina.



ROP Stages:

- **Stage 1** (Demarcation line separates vascular from avascular retina).
- **Stage 2** (Ridge formation with arteriovenous shunts).
- **Stage 3** (Ridge associated with extraretinal fibrovascular proliferation).
- **Stage 4** (Subtotal tractional RD).
- **Stage 5** (Total RD).
- **Treatment** for threshold Stage 3 is laser photocoagulation or cryotherapy to the avascular retina.



Retinal Vein Occlusions (CRVO & BRVO)

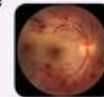
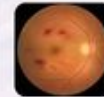
Central Retinal Vein Occlusion (CRVO)

Common in **old age**, **hypertension**, **diabetes**, and hyper-viscosity states. Arteriosclerotic changes in the central retinal artery **compress** the adjacent vein within the shared adventitial sheath.



CRVO Types:

- **1) Non-Ischemic:** Moderate visual drop, mild tortuous veins, dot-blot hemorrhages, few cotton wool spots, mild disc/macular edema, and good capillary perfusion on Fluorescein Angiography.
- **2) Ischemic:** Severe visual drop (<6/60), marked Afferent Pupillary Defect (RAPD), extensive dot-blot/flame hemorrhages in the periphery and center, numerous cotton wool spots, severe disc edema, and extensive non-perfusion areas. Complicated frequently by neovascularization of the iris (Rubeosis Iridis) leading to intractable neovascular glaucoma.



Branch Retinal Vein Occlusion (BRVO)

Dilated tortuous veins, hemorrhages, and edema strictly confined to the **wedge-shaped area** drained by the obstructed branch. The most common cause of persistent poor visual acuity here is **chronic macular edema**.



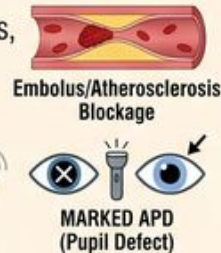
CENTRAL & BRANCH RETINAL ARTERY OCCLUSION (CRAO & BRAO): OCULAR EMERGENCIES & ISCHEMIA

CENTRAL RETINAL ARTERY OCCLUSION (CRAO)

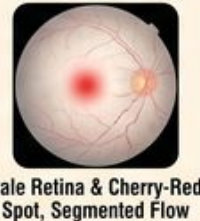
- **A TRUE OCULAR EMERGENCY!** Common causes include atherosclerosis, emboli (carotid/cardiac), and temporal arteritis.

CLINICAL PICTURE OF CRAO:

Presents with **SUDDEN, PAINLESS, PROFOUND**, and **DEVASTATING LOSS OF VISION** (patient can only see hand movement or less). A **MARKED** Afferent Pupillary Defect (APD) is instantly present.



- **FUNDUS SIGNS OF CRAO:** The entire retina becomes **OPAQUE, PALE**, and **MILKY-WHITE** (due to severe ischemic edema of the inner layers). A **CLASSIC "CHERRY-RED SPOT"** appears at the macula (because the thin fovea remains transparent, allowing the normal red choroidal circulation beneath to show through, contrasting sharply with the surrounding pale retina). Arteries are severely **ATTENUATED** with possible **"CATTLE-TRUCK"** segmentation of blood flow.



- **TREATMENT OF CRAO:** MUST be initiated **WITHIN 90 MINUTES** before irreversible retinal necrosis occurs. Methods include: **Lowering IOP** to **increase perfusion** (IV Acetazolamide, IV Mannitol, anterior chamber paracentesis), **Ocular massage** (to dislodge the embolus), **Sublingual nitroglycerin**, **Carbogen inhalation** (95% O₂ / 5% CO₂ to induce vasodilation), and **IV Streptokinase**.



BRANCH RETINAL ARTERY OCCLUSION (BRAO)

CAUSE & CLINICAL

PICTURE: Caused by emboli. Presents with a **SUDDEN, PAINLESS LOSS** of a **SPECIFIC SECTOR** of the visual field.



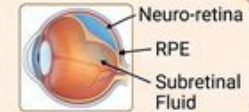
FUNDUS SIGNS OF BRAO:

Fundus shows a **LOCALIZED WEDGE-SHAPED AREA** of white ischemic retina corresponding **STRICTLY** to the distribution of the blocked arteriole.



RETINAL DETACHMENT (RD)

- **Definition:** Separation of the **sensory neuro-retina** from the underlying **retinal pigment epithelium (RPE)**. Fluid accumulates in the potential **subretinal** space.



1) RHEGMATOGENOUS RD

The **most common type**.

Caused by a **full-thickness retinal break** (tear or hole) allowing liquefied vitreous to gain access to the subretinal space.

Risk factors include **high myopia**, **aphakia/pseudophakia** (especially if complicated by vitreous loss), **severe trauma**, and **lattice degeneration**.

⚡ Symptoms

Flashes of light (photopsia) due to vitreoretinal traction, **floaters** (moving vitreous opacities), and a progressive **black curtain field defect**.

👁 Signs

Reduced IOP, grayish convex mobile undulating retina, and a visible reddish-pink retinal break. Treated surgically via Scleral Buckling or Pars Plana Vitrectomy with endolaser.



Liquefied Vitreous



2) TRACTIONAL RD

The sensory retina is mechanically pulled off by contracting **fibrovascular membranes** growing on the retinal surface.

Causes include **Proliferative Diabetic Retinopathy (PDR)**, **Sickle cell retinopathy**, severe **ROP**, and **penetrating trauma**.

(=) Signs

No primary retinal breaks, **smooth concave** retinal surface, **restricted retinal mobility**, and **no shifting fluid**. Treated by Pars plana vitrectomy and membrane peeling.



Fibrovascular Membranes



3) EXUDATIVE (SEROUS) RD

Fluid derived from **choroidal vessels** accumulates directly in the subretinal space without retinal breaks or traction.

🩸 Causes

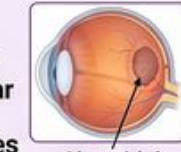
Choroidal tumors (malignant melanoma), severe **intraocular inflammation** (Harada's disease), or **systemic diseases** (toxemia of pregnancy, malignant hypertension).

💧 Signs

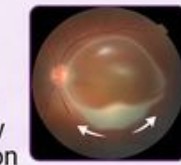
Bullous convex detachment, smooth surface, and highly characteristic **"shifting fluid"** (subretinal fluid moves rapidly to the most dependent position with changes in head posture).



Choroidal Vessels



Choroidal Tumor



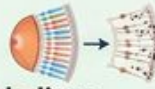
RETINITIS PIGMENTOSA (Primary Pigmentary Retinal Dystrophy)

DEFINITION

A **bilateral, hereditary, and slowly progressive primary degeneration** of the retinal photoreceptors (predominantly the **rods**) and the **retinal pigment epithelium**.

It is better termed "**pigmentary dystrophy**" due to the **complete absence of active inflammation**.

It can be **sporadic, autosomal dominant, autosomal recessive, or X-linked**.



CLINICAL SYMPTOMS

-  1) **Nyctalopia** (Night Blindness): The earliest and most prominent symptom, occurring due to the early, diffuse dysfunction and death of the peripheral **rod photoreceptors**.
-  2) **Progressive Loss of Peripheral Vision**: Leads to a severely constricted "**tubular**" visual field.
-  3) **Late Poor Central Vision & Color Vision**: Occurs when the **macular cone photoreceptors** eventually become involved in advanced stages.

CLASSIC CLINICAL TRIAD (Fundus Signs)

-  1) **Mid-peripheral pigment clumps** appearing classically as "**Bone-Spicules**" (representing migrating **RPE cells** into the sensory retina, gradually increasing in density and spreading anteriorly and posteriorly).
-  2) **Marked Arteriolar Attenuation** (thread-like retinal vessels).
-  3) **Waxy yellow pallor** of the **Optic Disc**.

ASSOCIATED OCULAR FEATURES

- **High myopia** 
- **open-angle glaucoma** 
- **posterior vitreous detachment**
- **posterior subcapsular cataract** 
- **keratoconus** 
- **cystoid macular edema** 

MANAGEMENT

No definite cure is currently known. Management focuses on treating associated complications (like **cataract extraction** , utilizing **low-vision aids** , and providing **genetic counseling** 

CHAPTER 12

Chapter 12: Neuro-ophthalmology

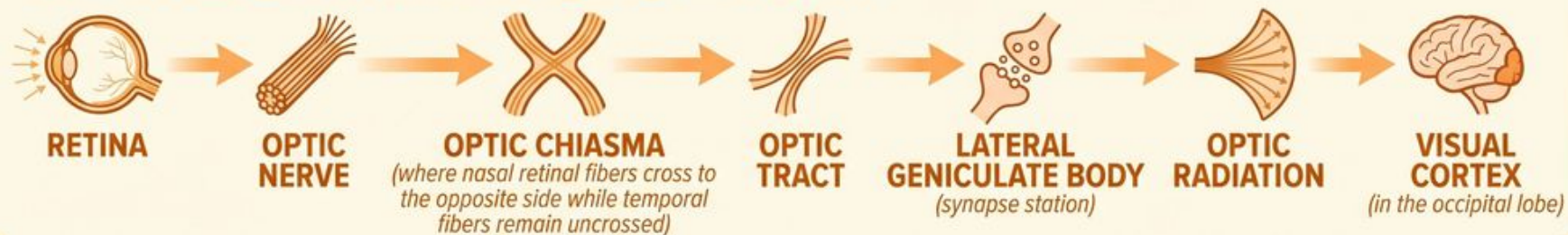


ANATOMY OF THE OPTIC NERVE AND VISUAL PATHWAY • PUPILLARY REFLEXES AND ABNORMALITIES • OPTIC NEURITIS (PAPILLITIS AND RETROBULBAR NEURITIS) •
PAPILLEDEMA (CHOKED DISC) • OPTIC ATROPHY (PRIMARY AND SECONDARY)

THE OPTIC NERVE: *Anatomy & Divisions*



THE VISUAL PATHWAY: *Light to Perception*



VISUAL FIELD DEFECTS: *Lesion-Based Loss*

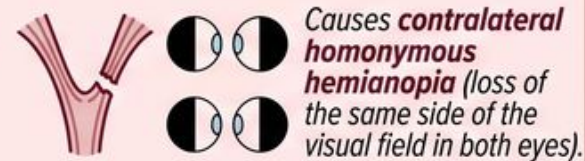
Optic Nerve Lesion:



Optic Chiasma Lesion (e.g., Pituitary adenoma):



Optic Tract or Radiation Lesion:

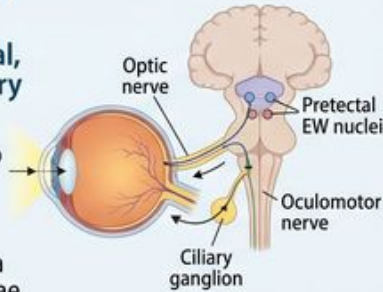


PUPILLARY REFLEXES AND ABNORMALITIES

1. LIGHT REFLEX PATHWAY

Afferent (Optic Nerve → Pretectal Nucleus), Bilateral Edinger-Westphal, Efferent (Oculomotor Nerve → Ciliary Ganglion → Sphincter Pupillae)

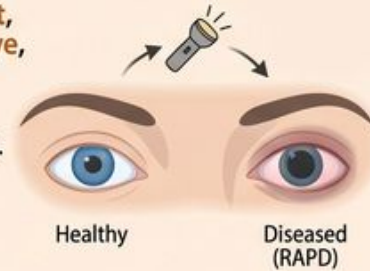
Afferent fibers travel via the optic nerve to the pretectal nucleus, sending bilateral fibers to both Edinger-Westphal nuclei. Efferent fibers travel via the Oculomotor (3rd) nerve to the ciliary ganglion, then via short ciliary nerves to the sphincter pupillae.



2. RAPD (MARCUS GUNN PUPIL)

Keypoints: Swinging Flashlight Test, Paradoxical Dilation in Diseased Eye, Unilateral Optic Nerve/Massive Retinal Disease

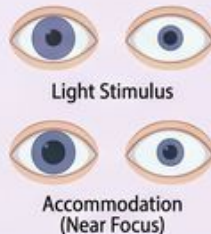
Detected by the swinging flashlight test. When light swings to the diseased eye, paradoxical pupillary dilation occurs. It indicates unilateral extensive optic nerve or massive retinal disease.



3. ARGYLL ROBERTSON PUPIL

Keypoints: Bilateral Small, Irregular Pupils, Light-Near Dissociation (constrict on accommodation, NO reaction to light), Neurosyphilis

Bilateral small, irregular pupils that demonstrate **Light-Near Dissociation** (they constrict on accommodation but completely fail to react to light). Classically pathognomonic for **neurosyphilis**.



4. ADIE'S TONIC PUPIL

Keypoints: Dilated Pupil, Extremely Sluggish Light Reaction, Slow Accommodation, Unilateral (Young Females)

A dilated pupil with an extremely sluggish reaction to light and slow accommodation, typically unilateral and seen in young females.



5. HORNER'S SYNDROME

Keypoints: Sympathetic Pathway Palsy, Triad: Miosis (constricted), Ptosis (drooping), Anhidrosis (no sweat)

A sympathetic pathway palsy presenting with the classic triad of **Miosis** (constricted pupil), mild **Ptosis** (drooping lid), and **Anhidrosis** (lack of sweating on one side of the face).



OPTIC NEURITIS: Papillitis and Retrobulbar Neuritis

Definition

An acute **inflammatory or demyelinating disease** of the optic nerve.



Inflamed Optic Nerve

Morphological Types

1) Papillitis

Inflammation involving the **intraocular portion (the optic disc)**.

More common in **children following viral infections**.



Viral Infections in Children



Swollen Optic Disc (Papillitis)

2) Retrobulbar Neuritis

Inflammation involving the **nerve behind the globe**.

More common in **adults** and strongly associated with **Multiple Sclerosis (MS)**.

Classically described as **"neither the doctor nor the patient sees anything"** because the **fundus appears initially normal**.



Initially Normal Fundus



Associated with MS

Clinical Picture

Sudden, **profound**, and **highly painful** loss of **unilateral** vision.

The **pain** is characteristically **exacerbated by eye movement**.



Pain with Movement

Clinical Signs

Marked **Relative Afferent Pupillary Defect (RAPD)**.



RAPD (Swinging Flashlight Test)

Visual field shows a **dense central scotoma**.



Central Scotoma

In papillitis, the fundus shows a **swollen, hyperemic disc with few hemorrhages** (differentiated from papilledema by the **severe visual loss and unilaterality**).



Papillitis Fundus

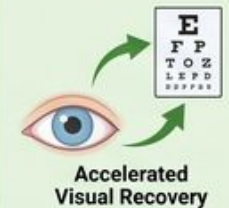


Papilledema Fundus

Differentiate by Severe Unilateral Loss

Treatment

High-dose **intravenous systemic corticosteroids (Methylprednisolone)** followed by an **oral taper** to **accelerate visual recovery**.



Papilledema (Choked Disc): An Explanatory Graphic

• Definition:

A **bilateral, non-inflammatory, passive swelling** of the optic disc secondary to **raised intracranial pressure (ICP)**.



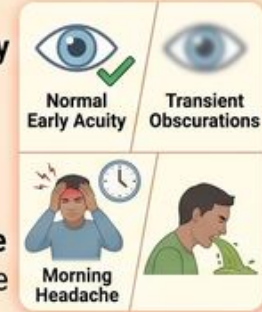
• Etiology:

Intracranial **space-occupying lesions** (brain tumors), **malignant hypertension**, **hydrocephalus**, and **benign intracranial hypertension** (pseudotumor cerebri).



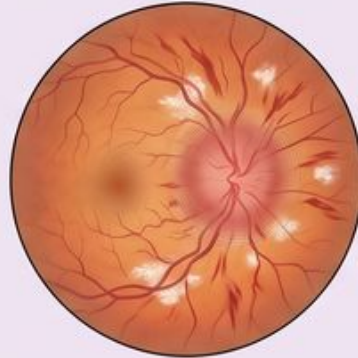
• Clinical Symptoms:

Visual acuity remains **remarkably completely normal** in the early stages, except for **transient visual** (seconds of blurring). Associated with **severe headache** (worse in the morning) and **projectile vomiting**.



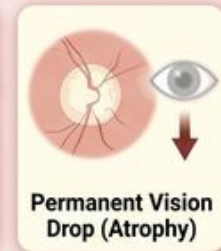
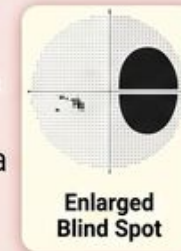
• Fundus Signs:

The optic disc is **hyperemic** and **markedly swollen** with **blurred margins**. The **physiological cup is obliterated**. The retina shows **flame-shaped hemorrhages**, **cotton wool spots** (infarcts), and **tortuous engorged veins**. **Paton's lines** (circumferential retinal folds) may appear.



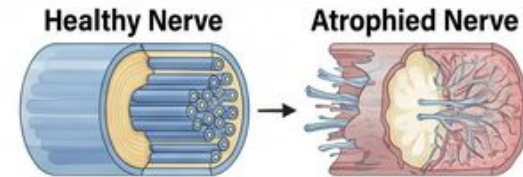
• Visual Field & Late Stage:

Displays an **enlarged blind spot**. If unrelieved, it progresses to **consecutive secondary optic atrophy**, resulting in a **permanent and profound drop in vision**.



OPTIC ATROPHY: Primary, Secondary, and Other Forms

Definition: Irreversible degeneration of optic nerve fibers with the loss of myelin and glial overgrowth, leading to a **pale optic disc** and **permanent visual loss**.



PRIMARY Optic Atrophy

Caused by **direct lesions to the nerve** (without prior swelling) like **trauma**, **toxic neuropathies** (e.g., methanol toxicity), or **compressive tumors**.

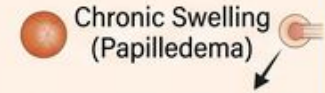


The fundus classically shows a **chalky white disc** with **sharply defined margins**, a **highly visible lamina cribrosa**, and completely **normal retinal vessels**.



SECONDARY Optic Atrophy

Follows **chronic, long-standing disc swelling** (e.g., chronic papilledema or papillitis).



The fundus shows a **dirty grey or pale disc** with **blurred, irregular margins**, an **obliterated hidden lamina cribrosa**, and **severely attenuated retinal vessels**.



Glaucomatous Atrophy

Characterized by **deep, wide cupping** with a **nasal shift** of the central vessels.



Consecutive Atrophy

Follows **widespread retinal destruction** (like **Retinitis Pigmentosa**), displaying a characteristic **waxy yellow disc pallor**.



CHAPTER 13

Chapter 13: Ocular trauma



CHEMICAL OCULAR BURNS (ACID AND ALKALI INJURIES) • PHYSICAL, THERMAL, AND RADIATION OCULAR INJURIES • THE EYELID INJURIES: ECCHYMOSIS, TRAUMATIC PTOSIS, & LACERATE • BLUNT INJURIES & FRACTURES OF THE ORBIT: RETROBULBAR • BLUNT OCULAR TRAUMA (ANTERIOR SEGMENT CONCUSSION) • BLUNT OCULAR TRAUMA (POSTERIOR SEGMENT AND ORBIT) • PENETRATING OCULAR INJURIES AND INTRAOCULAR FOREIGN BODIES

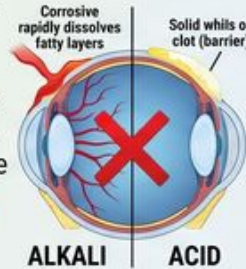
CHEMICAL OCULAR BURNS: ACID & ALKALI INJURIES (Explainer Graphic)

Mechanism of Injury: Acid vs. Alkali

Alkali burns (ammonia, lime, lye) are **far more dangerous and devastating** than acid burns.

Alkalis cause **rapid saponification** of cellular fatty acids, allowing them to **penetrate deeply and rapidly** into the

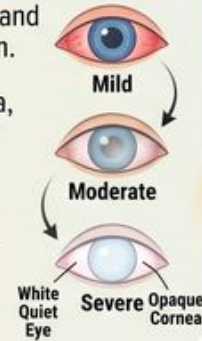
Acids cause **superficial coagulative protein necrosis**, which forms a **physical barrier that limits deeper penetration**.



Clinical Picture & Grading

Intense pain, photophobia, and blepharospasm are common.

Graded into **Mild** (hyperemia, red eye), **Moderate** (corneal haze, cloudy), and **Severe** (total **limbal ischemia** presenting as a '**white quiet eye**', and a totally **opaque porcelain cornea**).



Potential Complications

Severe Symblepharon (adhesion between palpebral and bulbar conjunctiva), **secondary glaucoma**, intractable **corneal ulceration**, perforation, and dense **vascularized leukoma**.



Emergency Management: IMMEDIATE ACTION

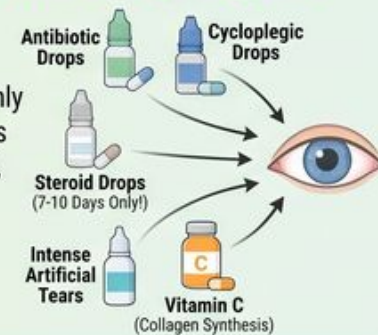
The **absolute priority** is **immediate, copious, and continuous irrigation** of the eye using any available neutral fluid (saline or tap water) for **at least 30 minutes** **BEFORE** taking any history or examination.



Double eversion of the eyelids is mandatory to sweep and remove any retained particulate matter (like lime chunks).

Medical Therapy & Recovery Support

Topical antibiotics, cycloplegics (to prevent synechiae), **topical steroids** (only in the **first 7-10 days** to suppress acute destructive inflammation), **intense artificial tears**, and **Vitamin C** (to promote **collagen synthesis** and prevent perforation).



III. Physical, Thermal, and Radiation Ocular Injuries

• 1. Thermal Burns

- **Causes:** Direct exposure to extreme heat sources, boiling liquids, steam, open flames, or flying molten metal droplets. 
- **Manifestations:** Causes severe **coagulative necrosis** of the eyelid skin, leading to massive **cicatricial contraction** and severe **cicatricial Ectropion or Lagophthalmos**. The cornea can display severe epithelial sloughing or deep **porcelain opacification**.
- **Management:** Copious lubrication with sterile ointments, systemic analgesics, local antibiotics, and early skin grafting to safeguard the cornea from **exposure keratitis**.

• 2. Ultraviolet (UV) Radiation Injury (Photokeratitis / Snow Blindness / Welder's Flash)

- **Causes:** Exposure to industrial electric welding arcs without protective goggles, or reflection of UV rays from snow surfaces or sunlamps. 
- **Pathogenesis:** The corneal epithelium strongly absorbs UV rays (specifically **UV-B**), leading to delayed, widespread desquamation and **punctate epithelial erosions**.
- **Clinical Picture:** A characteristic latency period of 6–12 hours is followed by the sudden onset of **agonizing ocular pain**, severe **foreign body sensation**, intense **photophobia**, reflex lacrimation, and severe **blepharospasm**.
- **Treatment:** Reassuring the patient that the condition is completely **self-limiting**. Managed with short-acting topical cycloplegics, broad-spectrum antibiotic ointments, and bilateral tight eye patch bandaging for 24 hours to allow rapid **epithelial regeneration**. Topical anesthetics are **strictly contraindicated** for home use.

• 3. Infrared (IR) Radiation Injury

- **Cause:** Prolonged, occupational exposure to intense heat and infrared rays over many years, classic in **glass-blowers** and furnace workers. 
- **Manifestations:** **True exfoliation** of the anterior lens capsule, where the superficial capsule peels off in thin ribbons into the anterior chamber, accompanied by the gradual development of a progressive posterior cortical lens opacity known as '**Glass-blower's Cataract**'.
- **Solar Retinopathy:** Looking directly at the sun (especially during an eclipse) focuses infrared and light rays onto the fovea, causing thermal and photochemical destruction of the **macular photoreceptors**, presenting with a **permanent central scotoma**.

I. The Eyelid Injuries: Ecchymosis, Traumatic Ptosis, & Lacerated Wounds

a. Ecchymosis (Black Eye)

- **Definition & Cause:** Extravasation of blood into the loose subcutaneous areolar tissue of the eyelids following blunt ocular or facial trauma.
- **Clinical Features:** Rapid swelling and bluish-purple discoloration of the eyelids.
- **Important Clinical Note:** If ecchymosis appears bilaterally without local trauma, it is highly pathognomonic for a **FRACTURE OF THE BASE OF THE SKULL** (known as 'Spectacle eyes' or 'Raccoon eyes').
- **Treatment:** **Immediate Cold Compresses/Ice Packs** in the first 24 hours to induce local vasoconstriction and limit bleeding, followed by **Warm Compresses** after 24 hours to accelerate blood absorption.



0-24 Hrs

>24 Hrs

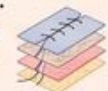
b. Traumatic Ptosis

- **Mechanism:** Drooping of the upper eyelid following injury, caused by **direct mechanical edema/hematoma**, **neuromuscular laceration of the Levator Palpebrae Superioris (LPS) muscle**, or a **traumatic lesion to the Oculomotor (3rd) nerve** or **Sympathetic fibers**.
- **Management:** **Edematous types** resolve spontaneously. **Neuromuscular lacerations** require **direct architectural primary micro-surgical repair**. **Paralytic types** require a **6-Month Observation Period** for potential nerve regeneration before attempting plastic correction.



c. Contused and Lacerated Wounds

- **Vertical/Oblique Lacerations:** **EXTREMELY DANGEROUS** because they cut across the horizontal fibers of the Orbicularis Oculi muscle, causing the wound edges to gape extensively due to muscle contraction, leading to **Severe Ectropion** or **Lid Coloboma**.
- **Canalicular Involvement:** Lacerations involving the **Medial 1/6th** of the lid margin can tear the lacrimal canaliculus, leading to **Permanent Structural Epiphora**.
- **Management:** Requires **Precise Primary Microsurgical Wound Closure** under a surgical microscope. **Horizontal lacerations** require normal skin sutures. **Vertical lacerations** must be sutured in **Multiple Layers (tarsus, muscle, skin)** to prevent lid margin notched deformities. **Lacerated canaliculi** require surgical repair over a **Silicone Stent** or **mini-Monoka tube** to maintain patency.



II. BLUNT INJURIES & FRACTURES OF THE ORBIT: Retrobulbar Hemorrhage, Ophthalmoplegia, & Orbital Wall Fractures

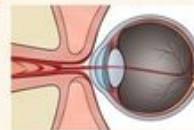
3. Orbital (Retrobulbar) Hemorrhage

Definition & Cause: Sudden, massive collection of blood within the closed retrobulbar orbital spaces, secondary to severe blunt trauma tearing the orbital or ciliary blood vessels.

Clinical Manifestations: Rapid onset of tense, non-compressible, severe axial proptosis, severe deep orbital pain, eyelid ecchymosis, subconjunctival hemorrhage extending without a visible posterior limit, and a marked relative afferent pupillary defect (RAPD).

Complications: The rapidly rising intraorbital pressure compresses the central retinal artery or the posterior ciliary nerves, leading to instant ischemic Optic Neuropathy or central retinal artery occlusion (CRAO), resulting in sudden permanent loss of vision.

Urgent Management: This is a true blinding emergency. Requires immediate Lateral Canthotomy and Inferior Cantholysis at the bedside to split the lid and instantly decompress the orbit. Accompanied by intravenous osmotic agents (Mannitol) and immediate surgical exploration if the intraocular pressure remains elevated.

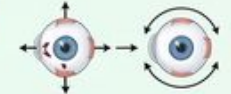


4. Post-Traumatic Ophthalmoplegia

Definition: Mechanical limitation or complete paralysis of extraocular muscle movements following trauma.

Causes: 1) **Mechanical Entrapment:** Herniation and trapping of an extraocular muscle (typically the Inferior Rectus) inside an orbital wall fracture line. 2) **Nerve Injury:** Traumatic palsy of the 3rd, 4th, or 6th cranial nerves. 3) **Local Muscle Damage:** Severe muscle contusion or retrobulbar hematoma.

Clinical Picture: Marked limitation of eye movement accompanied by severe, constant binocular diplopia.



5. Orbital Wall Fractures (Blow-out Fracture)

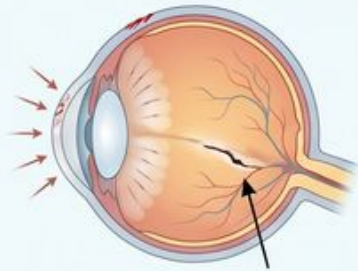
Mechanism: A blunt object larger than the orbital rim (such as a fist or tennis ball) strikes the orbit, causing a sudden, massive increase in intraorbital pressure. The force breaks the weakest, thinnest bony walls (the orbital floor over the maxillary sinus, or the medial wall into the ethmoid sinus), while the thick orbital rim remains completely intact.

Diagnostic Signs: Enophthalmos (the eyeball falls backward and downward into the maxillary sinus), Infraorbital nerve anesthesia (numbness of the cheek, upper lip, and teeth), vertical diplopia (double vision worse on up-gaze due to physical entrapment of the inferior rectus muscle), and localized orbital emphysema (crackling air under the skin upon blowing the nose due to communication with paranasal sinuses). Confirmed via urgent CT scan.



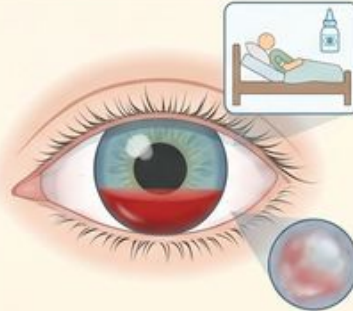
BLUNT OCULAR TRAUMA: ANTERIOR SEGMENT CONCUSSION

CORNEA



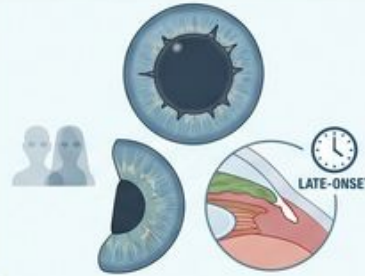
- Can present with **Epithelial Abrasions** (surface scratches), **Stromal Edema** (cloudiness from swelling), and **Endothelial Tears** (tears in the inner layer).

ANTERIOR CHAMBER (TRAUMATIC HYPHEMA)



- Collection of blood in the anterior chamber from **ruptured iris** or **ciliary body vessels**. Managed strictly of strictly by bed rest, head elevation, cycloplegics, and steroids. Major complications include **Secondary Glaucoma** and **Toxic Corneal Blood Staining**.

IRIS AND CILIARY BODY



- **Traumatic Mydriasis** (sphincter tears), **Iridodialysis** (tearing of the iris root forming a **D-shaped pupil** and causing **Monocular Diplopia**), and **Angle Recession** (tearing of the ciliary body face, leading to **Intractable Late-Onset Secondary Glaucoma**).

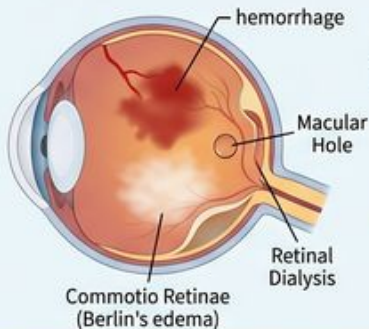
LENS



- **Vossius Ring** (a circular ring of **brown iris pigment imprinted** on the anterior capsule), **Rosette-shaped Concussion Cataract** (due to forced aqueous tracking along lens sutures), and **Lens Subluxation/Dislocation** due to the physical **Rupture of the Zonules of Zinn**.

Blunt Ocular Trauma: Posterior Segment & Orbit

Vitreous & Retina



Severe blunt impact causes **vitreous hemorrhage** from torn retinal vessels. Retinal manifestations include **Commotio Retinae** (Berlin's edema: a milky white retinal haze at the posterior pole), **traumatic macular holes**, **retinal dialysis** (tears at the ora serrata), and subsequent **Rhegmatogenous Retinal Detachment**.



Choroid

Traumatic choroidal rupture, classically appearing as a **crescent-shaped scar** concentric to the **optic disc**.



Traumatic Optic Neuropathy



Complete Avulsion



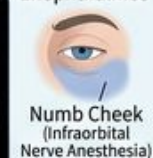
Optic Nerve

Traumatic optic neuropathy or complete **avulsion** from the globe, causing instant, **irreversible blindness**.

Orbit (Blow-out Fracture)



Enophthalmos



A sudden rise in intraorbital pressure fractures the **weakest bone** (usually the **orbital floor**). Clinical presentation includes **enophthalmos**, **infraorbital nerve anesthesia** (numb cheek), and **restricted up-gaze** causing **vertical diplopia** due to mechanical entrapment of the inferior rectus muscle in the fracture site.

Penetrating Ocular Injuries & Intraocular Foreign Bodies (IOFB)

1. Definition & Signs



Full-thickness sharp wounds to the globe.

- Full-thickness sharp wounds to the globe.
- Clinical signs include a **shallow or lost anterior chamber**, **prolapse** of uveal tissue (iris), a peaked or irregular pupil pointing towards the wound, **hypotony**, and **visible tissue lacerations**.



2. Intraocular Foreign Body (IOFB)



Diagnosis

Requires a careful slit-lamp exam, CT scan, and B-scan ultrasound.

Highly suspected in hammering, chiseling, or explosion injuries.



MRI is strictly contraindicated!
(Risk of moving a metallic body and shredding the globe).



3. Specific Metallic Reactions

Siderosis Bulbi (Iron)

Oxidation of an iron IOFB causes **rust-brown deposits** in the lens, iris **heterochromia**, and severe **retinal pigmentary degeneration** ending in permanent **blindness**.



Chalcosis (Copper)

A copper IOFB causes intense **purulent sterile inflammation** or deposits forming a **“sunflower cataract.”**



4. Management

Urgent surgical exploration, primary **meticulous wound closure**, **excision or reposition** of prolapsed uveal tissue, **broad-spectrum IV antibiotics**, **tetanus prophylaxis**, and IOFB removal via Pars Plana Vitrectomy.

